

simulator’s bias for maize is uncertain.

8 *Keywords:* Agrivoltaics, Light sharing, Solar tracking, Food–energy nexus,
9 Crop modelling, Differentiable simulation, Decision under uncertainty

10 **1. Introduction**

11 Two of the defining challenges of this century compete for the same re-
12 source: land. The energy transition requires terawatts of new solar pho-
13 tovoltaic (PV) capacity, and ground-mounted solar farms need roughly one
14 to two hectares per megawatt. Food production must rise while crops are
15 exposed to more frequent heat waves and droughts. Warming alone is pro-
16 jected to reduce global yields of maize, wheat, rice and soybean by several
17 per cent per degree (Zhao et al., 2017; IPCC, 2021), and a new generation
18 of climate and crop models shows these impacts emerging earlier than pre-
19 viously expected (Jägermeyr et al., 2021). Croplands are also where solar
20 power potential is greatest (Adeh et al., 2019). They are flat, sunny, acces-
21 sible and close to rural electricity demand, so converting farmland into solar
22 farms is tempting and controversial at the same time.

23 Agrivoltaic (AV) systems combine crops and PV modules on the same
24 land and aim to resolve this conflict (Dupraz et al., 2011; Dinesh and Pearce,
25 2016; Weselek et al., 2019). By sharing sunlight between modules and plants,
26 a hectare can deliver more food plus energy than the two uses side by side,
27 with land-equivalent ratios (LER) well above one in field trials and mod-
28 elling studies (Dupraz et al., 2011; Amaducci et al., 2018; Valle et al., 2017;
29 Trommsdorff et al., 2021). The partial shade also changes the crop micro-
30 climate. It lowers canopy temperature and evaporative demand, conserves
31 soil moisture and improves water-use efficiency, especially in drylands (Mar-
32 rou et al., 2013b; Hassanpour Adeh et al., 2018; Barron-Gafford et al., 2019;
33 Elamri et al., 2018; Weselek et al., 2021). Agrivoltaics is therefore a can-
34 didate adaptation measure against heat and drought stress, as reviewed by
35 Mamun et al. (2022), in addition to a mitigation measure through renew-
36 able electricity. The same shade, however, also reduces the light available for
37 photosynthesis. Regulators increasingly require AV systems to retain most
38 of the reference crop yield. The German pre-standard DIN SPEC 91434, for
39 example, requires at least 66% of the reference yield (DIN, 2021), and other
40 schemes and many farmers set stricter floors. How much electricity can be
41 produced without sacrificing food security depends on the crop, the local

42 climate, the year and the growth stage.

43 Most AV designs fix the geometry of the array, i.e. the ground coverage
44 ratio (GCR, module area per ground area) and the tilt or tracking rule, from
45 generic rules of thumb. Single-axis trackers add a degree of freedom: the
46 modules can be rotated away from the sun to let light through when the crop
47 needs it, as demonstrated for lettuce by Valle et al. (2017), or towards the
48 sun to maximise electricity. Choosing the rotation every day is a sequential
49 decision problem. The value of light depends on the crop’s phenological stage,
50 soil water and upcoming weather, and the effect of a decision appears in the
51 harvest months later. Recent work in this journal has used evolutionary
52 multi-objective optimisation to lay out PV arrays on farmland (Zhang et al.,
53 2026). Machine learning for crop management has mainly used reinforcement
54 learning with black-box simulators (Gautron et al., 2022), which would need
55 very many simulated seasons for global, multi-decadal assessments.

56 Two methodological strands are relevant. Differentiable process models,
57 which couple physical simulators with learning, allow parameters and poli-
58 cies to be fitted by gradient descent through the simulator (Reichstein et al.,
59 2019; Shen et al., 2023), and we use them for both calibration and control.
60 Decision making under deep uncertainty asks for designs that perform accept-
61 ably across plausible models and not only under the best estimate (Marchau
62 et al., 2019), and scenario-based designs have finite-sample guarantees un-
63 der convexity assumptions that we do not meet (Calafiore and Campi, 2006;
64 Campi and Garatti, 2008); our design rules are heuristic counterparts and
65 we compare them with a simple safety margin.

66 Two questions follow. First, how much of the electricity that an agri-
67 voltaic system can produce under a yield-retention floor comes from the array
68 density, and how much from dynamic light-sharing control of the trackers?
69 Second, how robust is that answer to the weakest part of any such study, the
70 crop response to shade, and does a controller learned from a simulator add
71 anything to a well-designed hand-written rule? Answering them requires
72 a simulator that couples the physics of light, PV, microclimate and crop
73 growth, a protocol in which all controllers face the same design constraints,
74 a check of the shade response against field data, and enough sites and seasons
75 to separate real effects from noise.

76 This paper makes three contributions. The first is a quantification of
77 light-sharing control under a yield floor. Light sharing is parameterised by
78 one daily level between full backtracking and the rotation that minimises the
79 shadow on the crop, and we compare static arrays, a seasonal schedule, a rule

that follows crop phenology, a rule that reacts to heat and drought stress, and a neural controller (SALS) trained through the simulator. The second is a history-based evaluation protocol. The array density of every site is chosen from its historical seasons only, with the same rule for every method, and performance is measured on held-out sites and years, over three spatial splits and three training seeds per split. The third is a differentiable agrivoltaic simulator that supports both, coupling hourly solar geometry, single-axis trackers with backtracking, row-to-row shading and PV electricity, the direct and diffuse light reaching the crop, canopy temperature and reference evapotranspiration under the modules, and the SIMPLE crop model (Zhao et al., 2019) extended with a flowering-stage shade penalty. All components are differentiable tensor operations (Karniadakis et al., 2021). The PV part is benchmarked against PVGIS. The crop response to shade is learned from the open data of a meta-analysis (Laub et al., 2022), checked against field trials on rice, wheat, potato and maize (Gonocruz et al., 2021; Weselek et al., 2021; Ramos-Fuentes et al., 2023), and propagated through the simulator as an ensemble.

We apply the protocol to 400 crop–site pairs that represent where maize, wheat, rice, soybean and potato are grown worldwide, sampled in proportion to their harvested area (Yu et al., 2020) with local crop calendars (Jägermeyr et al., 2021), and driven by NASA POWER daily weather for 2001–2019 (Sparks, 2018) under baseline and +1.5, +2 and +3 °C climates. The main findings are as follows.

1. Light sharing is the main lever on electricity at a given floor, although economically it pays only at high prices (item 6). At the same mean yield retention of 90%, arrays with light-sharing control carry 2.1 to 2.9 times the module area of static arrays and produce 1.8–2.4 times as much electricity (Section 3.1).
2. The best hand-written rule follows crop phenology and needs no weather forecast. Rules that react to heat or drought stress are not better and depend on forecast quality (Section S1).
3. The crop response to shade can be learned from data. A hierarchical model fitted to 428 observations from 58 field studies predicts a held-out study of the four staple crop types to about 0.22 in relative yield, no better than proportional loss (it is better only for crop types the simulator does not represent), so between-study spread is large. The simulator is close to these curves for cereals and potato but milder than

- the meta-analysis for maize (for soybean it lies within the interval of the data), whereas one maize trial is milder still; the gains for those two crops are therefore uncertain (Section 3.6).
4. A learned controller adds a small gain over the best rule: 0.044 LER over the written phenology rule but only 0.007 over a window optimised on the training seasons of each split. Training it on ensembles of shade responses, or giving it a farm-measurable biomass signal, does not help, and with the wider ensembles it did not retain its margin over the written rule in our training budget (Section 3.7).
 5. Compliance is sensitive to the uncertainty of the crop response that enters the design. A density chosen under the mean response leaves about 70% of crop-site pairs at the floor when another plausible response holds (oracle 85–87%). A rule built on the quantiles of an ensemble of responses raises this by 8 points for 0.03 LER, and a floor raised by 0.02 under the mean response gives no detectable difference at equal compliance (the intervals are too wide to establish equivalence); no rule is safe outside the ensemble (whose spread is mostly narrower than the between-study spread), so the ensemble is useful to size a margin (Section 3.7).
 6. Under illustrative costs, light sharing breaks even with the open field, at floor-designed densities, at 70 to 92 USD MWh⁻¹ (static arrays at 59), learning is worth at most about 0.3–0.4 kUSD ha⁻¹ yr⁻¹ over an optimised window at 90 USD MWh⁻¹ (the 95% intervals reach from about -0.1 to 0.8, and sensing and compute costs are excluded), and a conventional plant replacing the crop is more profitable above its break-even of 48 USD MWh⁻¹ (Section 3.8).
 7. At the yield floor, agrivoltaics on 5% of the world’s staple cropland could supply about 14 PWh yr⁻¹, but only with about 10 TWp of modules, several times the installed PV capacity of the world (Section 3.10).

2. Materials and methods

2.1. Global cropland sites, weather and crop calendars

We studied five staple crops that together cover about 670 Mha of harvested area and 580 Mha of physical cropland: wheat, rice, maize, soybean and potato. Harvested areas (rainfed and irrigated) were taken from the SPAM 2010 gridded production maps (Yu et al., 2020) and aggregated to 1°

cells. For each crop, 80 cells with at least 5000 ha of the crop were drawn without replacement with probability proportional to harvested area. An un-weighted average over the sample therefore approximates an area-weighted average over global production. The 400 crop-site pairs lie in 381 distinct cells in 50 countries (Fig. 2). Sowing and maturity dates came from the GGCM Phase 3 crop calendar at 0.5° (Jägermeyr et al., 2021). For wheat we used the winter-wheat calendar unless the coldest-month mean temperature was below -8°C , in which case the spring-wheat calendar was used. Daily weather for 2001–2020 was retrieved from NASA POWER (Sparks, 2018): global and diffuse shortwave irradiance, maximum, minimum and dew-point temperature, precipitation and wind speed at 2 m. Each site contributes 19 seasons (sowing years 2001–2019), each simulated as the 365 days after sowing, which gives 7600 site-seasons.

Climate change was represented by the global warming levels $+1.5$, $+2$ and $+3^\circ\text{C}$ relative to 2001–2020. Daily maximum, minimum and dew-point temperatures were shifted uniformly, which keeps relative humidity approximately constant and raises the vapour-pressure deficit. The atmospheric CO_2 concentration was set to 390, 450, 500 and 600 ppm for the baseline and the three warming levels. Precipitation and irradiance were left unchanged. This delta approach isolates the thermal signal of warming; its limitations are discussed in Section 4.5.

2.2. A differentiable agrivoltaic simulator

The simulator advances all site-seasons in parallel with a daily time step for the crop and an hourly time step for radiation and PV (Fig. 1). All equations are implemented as PyTorch tensor operations (Paszke et al., 2019), so that yields and electricity are differentiable with respect to the design and control variables.

Hourly irradiance. Daily global and diffuse irradiance were disaggregated into hourly values with the Collares-Pereira and Rabl (1979) and Liu and Jordan (1960) ratios and renormalised to the daily totals. Beam irradiance on the horizontal is the difference between the two. Solar position follows the standard declination and hour-angle equations.

Array geometry and PV electricity. We consider elevated horizontal single-axis trackers with north–south axes. The axis height is 1.25 module widths, e.g. 2.75 m for the common 2.2-m single-module-high (1P) tracker. The rotation ϕ is limited to $\pm 60^\circ$. The ground coverage ratio g is the module width

divided by the row pitch and is restricted to $g \leq 0.5$. For 2.2-m modules this keeps a pitch of at least 4.4 m, and a free strip of at least 2.2 m between the rows when the modules are horizontal, which is the stow position during field operations. Let ψ be the solar zenith angle projected onto the east–west vertical plane. The energy-optimal rotation with backtracking (Lorenzo et al., 2011; Marion and Dobos, 2013) is $\phi_{\text{bt}} = \psi - \text{sgn}(\psi) \arccos(\cos \psi / g)$ when $\cos \psi < g$, and $\phi_{\text{bt}} = \psi$ otherwise. The *light-sharing* rotation ϕ_{sh} is the feasible angle closest to edge-on to the sun, which minimises the shadow on the crop. Given a light-sharing level $u \in [0, 1]$ chosen once per day, the rotation in every hour is

$$\phi = (1 - u) \phi_{\text{bt}} + u \phi_{\text{sh}}. \quad (1)$$

Plane-of-array irradiance combines three terms: beam irradiance on the module, reduced by the fraction of the module shaded by the neighbouring row, $\max\{0, 1 - \cos \psi / (g |\cos(\phi - \psi)|)\}$; isotropic sky diffuse; and ground-reflected light (albedo 0.2). The cell temperature follows the NOCT model (45 °C), with an hourly air temperature interpolated sinusoidally between the daily extremes. Electricity per unit land is plane-of-array irradiance $\times \eta \times g$, with $\eta = 0.21 [1 - 0.0035(T_{\text{cell}} - 25)] (1 - 0.14)$ for module efficiency, temperature losses and system losses. The installed capacity is 2.1 g MWp ha⁻¹. A conventional ground-mounted PV plant ($g = 0.4$, backtracking, no crop) is the energy reference E_{PV} .

Validation of the PV model. We compared the simulated specific yield of an isolated tracker row ($g = 0.05$, negligible row-to-row shading) with PVGIS 5.2 (European Commission, Joint Research Centre, 2022), for a horizontal north–south single-axis tracker with 14% losses, at 30 randomly chosen sites (29 returned data). Across sites, the simulated and PVGIS specific yields were correlated with $r = 0.971$. The simulator was on average 4.9% lower (mean absolute deviation 5.5%). This is consistent with its $\pm 60^\circ$ rotation limit and with the different irradiance databases (NASA POWER vs. SARA2, ERA5 and NSRDB). The PVGIS benchmark covers an isolated row at low density only. Row-to-row shading and diffuse-light effects at higher ground coverage were compared with pvlib instead (Section 3.5); that comparison is model-to-model, used the same hourly irradiance and sun position, a rotation limit of 60° , an albedo of 0.2 and pvlib 0.16.1 (isotropic infinite-sheds model, no bifacial gain), and compares plane-of-array irradiance and not DC energy.

222 *Light and microclimate at the crop.* Averaged over the row pitch, the fraction
 223 of ground reached by beam light is $1 - \min\{1, g |\cos(\phi - \psi)| / \cos \psi\}$. The
 224 fraction of sky diffuse reaching the ground is a two-dimensional isotropic
 225 sky-view factor. We pre-computed it for 25 values of g and 13 rotation
 226 angles by ray casting over 31 neighbouring rows, and interpolate it bilinearly
 227 (differentiably). The daily ratio s of the irradiance at the crop to that of the
 228 open field defines the crop’s shading. The canopy maximum temperature is
 229 lowered by $\kappa(1 - s)$.

230 κ is uncertain. Field studies under PV modules report small reductions
 231 of daytime crop or canopy temperature and little change in air temperature
 232 (Marrou et al., 2013b; Barron-Gafford et al., 2019), and heat-stress thresholds
 233 calibrated on air temperature do not transfer directly to canopy temperature
 234 (Siebert et al., 2014; Webber et al., 2018). We therefore use a moderate
 235 $\kappa = 1^\circ\text{C}$ as the main case and report $\kappa = 0$ (no cooling) and $\kappa = 2^\circ\text{C}$
 236 for all methods (Section 3.2). Nocturnal warming under the modules is not
 237 represented. The reference evapotranspiration under the modules, ET_0 , is re-
 238 computed with the FAO-56 Penman–Monteith equation (Allen et al., 1998)
 239 from the reduced radiation and canopy temperature. Soil water conserva-
 240 tion under the modules (Hassanpour Adeg et al., 2018; Elamri et al., 2018)
 241 therefore emerges from the model, but rain redistribution by the modules
 242 (Marrou et al., 2013a) is ignored.

243 *Crop growth.* Crop growth follows the SIMPLE model (Zhao et al., 2019)
 244 with the species parameters of its Table 1a (Table 1). Daily biomass growth
 245 is

$$\Delta B = R_{\text{crop}} \cdot \text{RUE} \cdot f_{\text{Solar}} \cdot f_{\text{CO}_2} \cdot f_{\text{Temp}} \cdot \min\{f_{\text{Heat}}, f_{\text{Water}}\}, \quad (2)$$

246 where $R_{\text{crop}} = s R$ is the radiation received by the crop. The terms are as
 247 follows.

- 248 • f_{Solar} is the canopy’s radiation interception. In SIMPLE it depends
 249 only on thermal time, with senescence accelerated by heat and drought
 250 stress. Shade therefore reduces the daily biomass increment but not
 251 leaf-area development.
- 252 • f_{Heat} decreases linearly between the heat threshold T_{heat} and the ex-
 253 treme temperature T_{ext} of the canopy maximum temperature.
- 254 • $f_{\text{Water}} = 1 - S_{\text{water}} \text{ARID}$, where the agricultural reference index for
 255 drought (Woli et al., 2012) is computed from a daily soil-water balance

256 with curve-number runoff, drainage and root water uptake. Irrigated
 257 fields have $f_{\text{Water}} = 1$.

258 SIMPLE has a fixed harvest index and therefore ignores the high sen-
 259 sitivity of grain number to shading around flowering (Fischer, 1985; An-
 260 drade et al., 1999). It also ignores the differences in shade tolerance be-
 261 tween crop types. In the meta-analysis of Laub et al. (2022), maize and
 262 grain legumes lost yield strongly even at low shade levels, whereas C3 ce-
 263 reals and tuber crops initially lost less than proportionally; shading trials
 264 also show marked effects of shade on maize development (Ramos-Fuentes
 265 et al., 2023). We added a stage-dependent penalty. During a critical window
 266 ($0.45 \leq \text{TT}/T_{\text{sum}} \leq 0.65$), the harvest index is multiplied by $1 - h\bar{d}$, where $\bar{d}$
 267 is the mean relative radiation deficit $1 - s$ in that window. The sensitivity h
 268 is highest for maize (0.6) and soybean (0.5), the crop types that Laub et al.
 269 (2022) found most shade-sensitive, lower for the C3 cereals (wheat 0.4; rice
 270 0.5, for which no specific curve was available) and lowest for potato (0.2).
 271 These values are assumptions that reproduce this ranking; they are not cal-
 272 ibrated estimates. Section 3.5 shows the resulting shade-response curves.
 273 Because RUE is linear, SIMPLE does not credit the higher light-use effi-
 274 ciency of shaded canopies under more diffuse light. The penalty, conversely,
 275 makes shading around flowering more damaging than proportional. The net
 276 effect is closer to the field evidence than either assumption alone.

277 The thermal-time requirement to maturity was calibrated for every site as
 278 the mean thermal time between the GGCM sowing and maturity dates under
 279 the 2001–2020 climate, and the canopy parameters were scaled accordingly.
 280 The cultivar is thus adapted to the present climate, and warming shortens the
 281 season. Growth stops smoothly once the thermal-time requirement is met.
 282 Yield is biomass multiplied by the (shade-adjusted) harvest index. Each site
 283 is represented by a rainfed and an irrigated sub-field in proportion to its
 284 SPAM irrigated fraction. Each sub-field carries its own array, so both yield
 285 and electricity are area-weighted over the two sub-fields.

286 2.3. SALS: simulation-trained adaptive light sharing

287 SALS learns a *control network* $u_t = \pi_{\omega}(\mathbf{x}_t; g)$ (Algorithm 1). It is a
 288 multilayer perceptron with two hidden layers of 64 units and a bounded-logit
 289 output, $u = \sigma(5 \tanh(z/5)) \in [0.007, 0.993]$, which keeps gradients non-zero
 290 at the extremes. It chooses the daily light-sharing level of Eq. (1). Its inputs
 291 are the information available on day t :

- the crop and soil state: relative thermal time TT/T_{sum} , radiation interception f_{Solar} , relative soil water and the previous day’s ARID;
- a *forecast* of the day’s maximum temperature (relative to T_{heat} and T_{opt}), irradiance and ET_0 ;
- static information: the irrigation status of the sub-field, the array’s GCR g , latitude, day of season and the crop.

Forecasts are perturbed with random errors, independently every day: Gaussian with a standard deviation of 2°C for maximum temperature, and multiplicative with standard deviations of 20% for irradiance and 15% for ET_0 . The physics always uses the true weather. The same forecast errors are applied to the rule-based controllers. The controller is conditioned on g . During training, each season is simulated with a GCR close to the density that is sustainable at its training site. This is the largest value on a coarse grid $(0.05, 0.10, \dots, 0.50)$ at which seasonal sharing keeps the mean retention of the site’s training seasons at or above ρ , plus a uniform perturbation in $[-0.05, +0.10]$ truncated to $[0.05, 0.50]$. The network therefore learns to control the densities that design selection will later pick. Outside the cropping season the trackers backtrack for maximum electricity.

For site-season i with open-field yield Y_i^{open} , let $r_i = Y_i/Y_i^{\text{open}}$ be the yield retention and $e_i = E_i/E_{\text{PV},i}$ the electricity relative to a conventional PV plant. Training minimises the penalised objective

$$\mathcal{L}(\omega) = \mathbb{E}_{i \sim \mathcal{B}} \left[-e_i + \mu_1 \max\{0, \rho - r_i\} + \mu_2 \max\{0, \rho - r_i\}^2 \right], \quad \rho = 0.9, \mu_1 = 5, \mu_2 = 30, \quad (3)$$

which combines an exact (hinge) and a quadratic penalty for the yield-retention floor ρ . Because r_i and e_i are outputs of the differentiable simulator, the exact gradient of Eq. (3) is obtained by back-propagation through the 365 daily steps of every season (back-propagation through time). This is the key difference from reinforcement learning with black-box simulators (Gautron et al., 2022), which estimates gradients from sampled returns.

Each minibatch contains 512 site-seasons, each simulated as a rainfed and an irrigated sub-field, drawn from one of the four climates in turn. Seasons with a failed open-field crop ($Y^{\text{open}} < 0.2 \text{ t ha}^{-1}$) are excluded from training and evaluation. The network is fine-tuned through the simulator with Adam (Kingma and Ba, 2015) (learning rate 10^{-3} , cosine decay, gradient-norm clipping at 1) for 300 steps. Growth-stage logistic functions are evaluated in

Algorithm 1 SALS training

Require: training site-seasons in four climates; floor ρ ; penalties μ_1, μ_2

- 1: initialise π_ω by behaviour cloning of the phenology rule
- 2: **for** step = $1, \dots, N$ **do**
- 3: sample a climate and a minibatch $\mathcal{B}$ of site-seasons (rainfed and irrigated sub-fields)
- 4: set g_i to the site’s sustainable density plus a perturbation; initialise crop and soil states
- 5: **for** day $t = 1, \dots, 365$ **do**
- 6: $u_{it} \leftarrow \pi_\omega(\mathbf{x}_{it}; g_i)$ from state and noisy forecast if the crop is in the field, else 0
- 7: hourly rotation (Eq. 1), PV electricity, crop light, canopy temperature, ET_0
- 8: advance the soil-water balance and the SIMPLE crop state (Eq. 2)
- 9: $r_i \leftarrow Y_i/Y_i^{\text{open}}, e_i \leftarrow E_i/E_{\text{PV},i}$ (area-weighted); loss $\mathcal{L}$ (Eq. 3)
- 10: back-propagate through the season; Adam update of ω

325 a numerically stable form, so that gradients remain finite even under extreme
326 heat.

327 *Initialisation.* Trained from random weights, the controller converged to
328 sharing light throughout the cropping window ($u > 0.99$ on all days), i.e. to
329 the seasonal-sharing rule, because the gradient of electricity with respect to u
330 is small and of the same sign for most days of a season. We therefore initialise
331 the network by behaviour cloning of the phenology rule of Section 2.4: binary
332 cross-entropy against the rule’s output (soft targets 0.95 and 0.05) on about
333 1.7 million day-states visited by the rule in the training seasons, with 1500
334 Adam steps at a learning rate of 3×10^{-3} . Fine-tuning then moves the
335 policy away from this initialisation wherever doing so improves the objective.
336 Consequently, the gain of SALS over the phenology rule measures what fine-
337 tuning through the simulator adds to that rule, not what can be learned
338 without it.

339 2.4. Deployable design selection and baselines

340 A real array has one fixed density, which must be chosen before it is built.
341 Every method, SALS included, therefore uses the same *history-based design*
342 *rule*. For a site, the GCR is the largest value on the grid $\{0.05, 0.075, \dots, 0.50\}$
343 for which the method’s mean yield retention over the site’s 2001–2014 sea-

sons is at least $\rho = 0.9$. If no value qualifies, no array is built and the site remains open field. A stricter *chance-constrained* variant requires $r \geq 0.9$ in at least 80% of the historical seasons. The historical seasons of the test sites are never used for training, and the weather of the evaluation years is never used for design. We compare four controllers:

- *AV static*: backtracking trackers without light sharing ($u = 0$);
- *Seasonal sharing*: the light-sharing rotation ($u = 1$) throughout the cropping season;
- *Stress rule*: an expert rule that shares light ($u = 1$) during the main growth phase ($0.1 < TT/T_{\text{sum}} < 0.9$) except on days with a heat-stress forecast ($T_{\text{max}} \geq T_{\text{heat}}$) or drought ($\text{ARID} > 0.5$), when the modules track the sun and shade the crop;
- *SALS*: the learned controller.

The open field and a conventional PV plant without crops ($\text{LER} = 1$ each) are the references. Because the design rule is identical for all controllers, differences between methods reflect the quality of control and the array density that each controller can sustain under the yield-retention floor. We also compare the controllers at identical GCR over the whole grid (matched-density frontiers).

2.5. A shade-response model learned from field data

The simulator needs a crop response to shade that is supported by observations. We learned one from the open data of the meta-analysis of Laub et al. (2022) (Zenodo 10.5281/zenodo.5716091, CC BY 4.0): 428 observations of yield relative to an unshaded control at relative shade reductions of 0–90% from 58 studies, with crop and crop type, experiment type (artificial shading or intercropping), climate zone, irradiation and shade type. Relative yield is modelled as $y(s) = \exp(-\beta s^\gamma)$ with Student- t errors ($\nu = 4$; maximum a posteriori estimation, so this is a hierarchical shrinkage regression with bootstrap intervals and not a full Bayesian model). We compared six models: proportional loss $y = 1 - s$; one curve for all crops; one curve per crop type without pooling; one curve per crop type with partial pooling (Gaussian shrinkage of $\log \beta$ and $\log \gamma$ towards a common value, scale 0.5); partial pooling with covariates on $\log \beta$ (standardised irradiation, subtropical zone, solar-panel shade and intercropping); and gradient-boosted trees

on the same inputs. Fits use L-BFGS. Models were compared by leave-one-study-out cross-validation, in which every study is held out once, which tests the prediction of a new study and not of a new observation in a known study. Intervals for the crop-type curves and covariate effects come from a cluster bootstrap over studies (150 resamples).

2.6. Shade-response uncertainty: gradient calibration, robust controllers and design

The crop’s response to shade is the least certain part of the simulator, so we treat it as an unknown with a distribution. Two parameters per crop enter the simulator: a radiation-use compensation c (growth is multiplied by $1 + c(1 - s)$, s being the share of open-field irradiance that reaches the crop) and a multiplier h on the flowering-window harvest-index penalty. Because the simulator is differentiable, both can be fitted by gradient descent through whole simulated seasons to published yield–shading observations. The observations are the relative yields at 20% and 40% shading of the data-driven curves of Section 2.5 for the five crops (median, with an error equal to the width of the 80% bootstrap interval divided by 2.56 and at least 0.04; rice takes the curve of C3 cereals, which contains no rice), the field trials on rice, wheat, potato and maize (Gonocruz et al., 2021; Weselek et al., 2021; Ramos-Fuentes et al., 2023) (each season of each trial as a separate observation, error 0.06–0.08) and the unaffected harvest index of shaded maize (Ramos-Fuentes et al., 2023). We call the resulting ensemble E2. A wider ensemble, E3, uses the same procedure with a target error of 0.2 for the learned curves, which is the between-study spread of a new study (Section 3.6); the spread of the resulting members is smaller than that error (standard deviation of relative yield across members 0.02–0.09 at 20% shading and 0.03–0.15 at 40% shading, against 0.02–0.04 and 0.02–0.06 for E2, depending on the crop; Supplementary data, ensemble spread), because the fit also weights the field trials and the priors. An earlier ensemble, E1, was fitted in the same way to curves read by hand from the figures of Laub et al. (2022) and was used to train the robust and adaptive controllers below. Priors are $c \sim \mathcal{N}(0.4, 0.5^2)$ and $h \sim \mathcal{N}(1, 0.5^2)$, and the parameters are clamped to $[0, 2.5]$ (c) and $[0, 2]$ (h) after every step, which is why a response milder than the data cannot be lowered further. We fitted 12 ensemble members by randomised maximum a posteriori estimation: every member sees the observations perturbed by their errors and a prior draw perturbed by its width, and is optimised with Adam (40 steps, with the loss still falling slightly at the last step; 40 training-block

415 seasons per crop and member; shading of the static arrays cached). The
 416 ensemble is a device for generating plausible responses and not a validated
 417 posterior: it has no coverage check, and for some parameters (the harvest-
 418 index multiplier of wheat and potato) its spread equals that of the prior.
 419 The design analyses use five fixed members (indices 0, 3, 5, 8 and 10 of 12),
 420 chosen in advance. The ensemble is used in three ways. All training and
 421 evaluation ran on a 4-core CPU; one training run takes about 20–25 minutes
 422 and the calibration of one ensemble about 7 minutes.

423 *Tuned rules.* The written phenology rule shares light while thermal time
 424 lies between 0.05 and 1.0 of the season requirement. To avoid comparing
 425 a trained controller with an untuned rule, we tuned this window on the
 426 training-block seasons of each split (500 random seasons, baseline climate,
 427 main shade response) with the objective used for all rules: the relative elec-
 428 tricity at the density where the mean retention equals the floor, found by in-
 429 terpolation over five densities (0.1 to 0.5). A 4×3 grid of start (0, 0.05, 0.15,
 430 0.3) and end (0.6, 0.8, 1.0) gave the grid-tuned window. A five-parameter
 431 schedule (start in $[0, 0.6]$, end in $[0.5, 1.2]$, two ramp widths and a level in
 432 $[0, 1]$, all through logistic or exponential transforms) was then optimised by
 433 CMA-ES (Hansen and Ostermeier, 2001) with 300 seasons, population 8 and
 434 25 generations (200 evaluations) per run; run 1 started at the grid-tuned win-
 435 dow and runs 2 and 3 at random points, and the run with the best training
 436 objective was evaluated on that split (the CMA-ES schedule and the grid
 437 window were tuned separately on each split’s training block; on split 0 run 2
 438 was best, on split 1 run 2 and on split 2 run 1). The grid window was 0.15 to
 439 1.0 on all three splits, and the schedules of the best runs had starts of 0.20,
 440 0.20 and 0.22, ends of 0.93, 0.92 and 0.93, a level of 1.0 and ramp widths
 441 below 0.01 of the requirement. On split 0 run 3 stopped in a worse optimum
 442 (objective 0.345 against 0.377) and on split 1 as well (0.349 against 0.379),
 443 so three runs give no guarantee of global optimality.

444 *Robust and adaptive controllers.* SALS-robust has the architecture and
 445 training of SALS (Section 2.3), but every training season is simulated with the
 446 response of a randomly drawn ensemble member that the controller does not
 447 observe, and the density jitter is widened to $[-0.05, +0.25]$ because feasible
 448 densities differ between members. SALS-adaptive additionally observes the
 449 realised biomass of the crop under the modules relative to an open reference
 450 plot, a quantity that can be measured on the farm by remote sensing or by
 451 a reference plot. A feedback rule that shares light whenever this ratio falls
 452 below a target serves as the hand-written counterpart; its target (1.0) was

453 chosen on training-block seasons under the ensemble.

454 *Uncertainty-aware design.* The design rule of Section 2.4 chooses the
455 largest GCR whose historical retention meets the floor. Under shade-response
456 uncertainty we choose the largest GCR that meets the floor in at least 75%
457 of the ensemble members (three of four in the leave-one-out analysis, four
458 of five otherwise), and compare it with a design under the ensemble mean
459 and with an oracle design under the true response. Evaluation of this rule
460 is leave-one-out: the truth is the held-out member and the design uses the
461 other four.

462 *Design rules compared.* With the ensemble members indexed k and one
463 member held out as the truth, the density of a pair is chosen from its 2001–
464 2014 seasons as (i) the *oracle*: the largest GCR whose mean retention under
465 the truth meets the floor; (ii) the *mean-response design*: the same rule under
466 the mean response of the other four members (the mean of their parameters);
467 (iii) a *margin*: the same rule under that mean response with the floor raised
468 to 0.92, 0.94 or 0.96; and (iv) the *ensemble rule*: the largest GCR at which
469 the mean retention under at least a share q of the other four members meets
470 the floor ($q = 0.5$: two of four; 0.75 : three of four; all: four of four). Out
471 of the ensemble, the five members and their mean (the mean of all twelve
472 members for the mean-response design) are used in the same way and the
473 truth is one of the four fixed responses below. The margin grid was chosen
474 after a first analysis of one ensemble and is not pre-specified.

475 *Stress tests.* Controllers and rules are also evaluated under four fixed
476 responses, the proportional response, the response calibrated to the meta-
477 analysis, the response fitted to the field trials and a harsh response (propor-
478 tional growth and a 1.5 times larger harvest-index penalty). The proportional
479 and harsh responses lie outside the range of the ensemble.

480 2.7. Economic analysis

481 To judge whether the electricity gains matter economically, we computed
482 an annual net value per hectare for each method relative to the open field:
483 electricity revenue minus the annualised capital cost (annuity) and operating
484 cost of the installed capacity minus the loss of crop revenue (crop produc-
485 tion costs are unchanged). The installed capacity is 2.1 g MWp per hectare
486 (modules cover a share g of the land at 0.21 kWp m^{-2}). Yield is dry matter,
487 valued at assumed prices. A conventional ground-mounted plant (GCR 0.4)
488 that replaces the crop is the comparator. All parameters (Table 9) are as-
489 sumptions chosen to be representative, not data, and their ranges drive the

490 conclusions; we therefore report break-even electricity prices and one-at-a-
 491 time sensitivities, and we do not monetise food security, which market crop
 492 prices do not capture. The analysis omits land rent, module degradation,
 493 the extra structure cost of sparse arrays, curtailment, subsidies and the un-
 494 certainty of the parameters themselves, which the bootstrap intervals do not
 495 include.

496 2.8. Evaluation protocol, statistics and upscaling

497 Sites were split spatially: $10^\circ \times 10^\circ$ blocks were randomly assigned to train-
 498 ing (70%) or test (30%), so that no test site has a training site in the same
 499 block. Training uses the 2001–2014 seasons of the training sites. Design
 500 selection uses the 2001–2014 seasons of the test sites. Evaluation uses the
 501 2015–2019 seasons of the test sites. The same site-seasons are analysed in all
 502 four climates.

503 We report the following metrics:

- 504 • the mean electricity ($\text{MWh ha}^{-1} \text{yr}^{-1}$) and the relative electricity e ;
- 505 • the mean yield retention r and the production-weighted retention $\sum Y / \sum Y^{\text{open}}$;
- 506 • season-level compliance (share of seasons with $r \geq \rho$) and site-level
 507 compliance (share of sites whose mean test-season retention is $\geq \rho$);
- 508 • the lower tail (10th percentile of r , share of seasons with $r < 0.8$);
- 509 • the land-equivalent ratio $\text{LER} = r + e$ (Dupraz et al., 2011), which gives
 510 equal weight to the retained yield and to the electricity relative to a
 511 conventional plant, and is relative to the open field of the same climate
 512 (a higher LER under warming reflects the lower reference electricity
 513 share and not a benefit of agrivoltaics).

514 Two aggregations are used and labelled in the text: tables report means over
 515 all test seasons of a sample (season-pooled), whereas statistical comparisons
 516 and intervals use the crop–site pair as the unit of analysis (mean of the pair’s
 517 seasons, then mean over pairs), so the same contrast can differ slightly be-
 518 tween the two (for example 0.034 and 0.038 for SALS minus the phenology
 519 rule on the main split). Samples are also labelled: the main analysis uses
 520 the 92 held-out pairs of one split and a GCR grid with step 0.025; analyses
 521 pooled over three splits use the 76 distinct pairs of every third held-out pair
 522 and a grid with step 0.05, and analyses of alternative shade responses on

Table 1: SIMPLE crop parameters (Zhao et al., 2019, Table 1a) and the harvest-index shade sensitivity h added in this study. T_{sum} , I_{50A} and I_{50B} are rescaled per site to the local crop calendar (Section 2.2). Units: temperatures in °C, thermal times in °C d, RUE in gMJ⁻¹; S_{CO_2} is the relative RUE increase per 100 ppm above 350 ppm.

Crop (cultivar)	h	T_{sum}	HI	I_{50A}	I_{50B}	T_{base}	T_{opt}	RUE	$I_{50 \text{ max } H}$	$I_{50 \text{ max } W}$	T_{heat}	T_{ext}	S_{CO_2}	S_{water}
Wheat (Batten)	0.4	2150	0.34	280	50	0	15	1.24	100	25	34	45	0.08	0.4
Rice (IR72)	0.5	2300	0.47	850	200	9	26	1.24	100	10	34	50	0.08	1.0
Maize (McCurdy 84aa)	0.6	2050	0.50	500	50	8	28	2.10	100	12	34	50	0.01	1.2
Soybean (Williams 82)	0.5	2350	0.40	600	200	6	27	0.86	120	20	36	50	0.07	0.9
Potato (Sebago)	0.2	2400	0.85	500	350	4	22	1.30	50	30	34	45	0.10	0.4

one split use every sixth (21 pairs) or every fourth pair (22 pairs). Statistical comparisons use site means as the unit of analysis. We report paired Wilcoxon signed-rank tests and cluster (site) bootstrap 95% confidence intervals of the mean difference. SALS was trained with three random seeds on each of the three spatial splits; the controller comparisons pooled over splits use the mean of the three seeds, and the economic analysis uses seed 0.

For the global estimate, every site is evaluated in the split in which it is held out. The mean electricity and yield per hectare of each crop are scaled to the *physical* area of that crop in SPAM 2010, which avoids double counting multiple-cropped land (Waha et al., 2020). We report the installed capacity (TWp), the capacity factor and a supply curve.

3. Results

3.1. Electricity and food on held-out sites and years

Table 2 compares the four controllers on the 92 held-out crop–site pairs of the main split (95 pairs are held out; pairs without a season whose open-field yield is at least 0.2 t ha⁻¹ in all four climates are excluded) (seasons sown in 2015–2019), after each site’s density has been chosen from its 2001–2014 seasons with the same rule for all controllers. By construction all methods keep the mean yield retention close to 90% (0.904–0.917), so differences in electricity are differences in the density that each controller can sustain under this floor.

Static arrays are the weakest. Without light sharing, the floor limits the ground coverage to 6% on average, and arrays produce 217 MWh ha⁻¹ yr⁻¹, 16% of a conventional PV plant per hectare (LER 1.07). Rotating the modules to let light through during the cropping season (seasonal sharing) allows a coverage of 17% and 397 MWh ha⁻¹ yr⁻¹ (LER 1.19). A rule

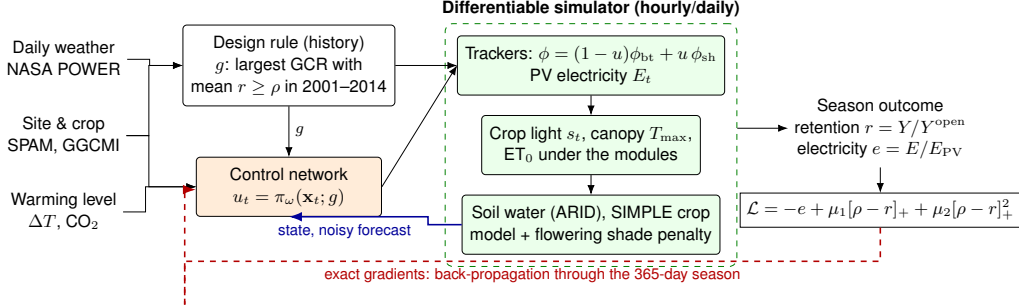


Figure 1: Simulator and controller. The array density g of a site is fixed from its historical seasons. A control network maps the daily crop and soil state, the density g and a noisy forecast to the light-sharing level u_t . The differentiable simulator converts these decisions into PV electricity, light and microclimate at the crop, soil water and crop growth. The season’s electricity and yield retention enter a penalised objective whose exact gradient is back-propagated through the 365-day season.

that shares light only while the crop is growing (phenology rule) reaches 465 MWh ha⁻¹ yr⁻¹ (LER 1.24). An expert rule that additionally shades the crop on forecast heat or drought days (stress rule) is worse than the phenology rule (LER 1.18): its density is limited to 13% because it reduces light at times when the floor is most binding.

The learned controller is best by a modest margin over the written rule. SALS sustains a density of 17% (similar to the phenology rule), produces 510 MWh ha⁻¹ yr⁻¹ and reaches an LER of 1.270 at the same retention. Compared with the phenology rule this is +0.038 in LER (95% CI over sites 0.025–0.053; +10% electricity) and it is higher at 84% of the 92 sites (Table S3). Compared with static arrays it is +0.205 (0.176–0.236), with 2.35 times the electricity at 2.9 times the density. The frontier in Fig. 3 shows the same picture at matched densities: the learned policy lies on or slightly above the phenology rule over the whole density range, both lie well above seasonal sharing and the stress rule, and static arrays are far below.

What the floor does and does not guarantee. Enforcing the floor on the historical mean does not guarantee it in every season. In the test seasons, 70% of SALS seasons have $r \geq 0.9$ (phenology rule 69%, stress rule 76%, static arrays 93%), and 2.7% have $r < 0.8$ (1.3%, 0.9% and 0.9%). The lower tail of SALS is therefore slightly heavier than that of the rules: its 10th percentile of r is 0.883, against 0.887, 0.886 and 0.903. Requiring $r \geq 0.9$ in at least 80% of a site’s historical seasons (chance-constrained

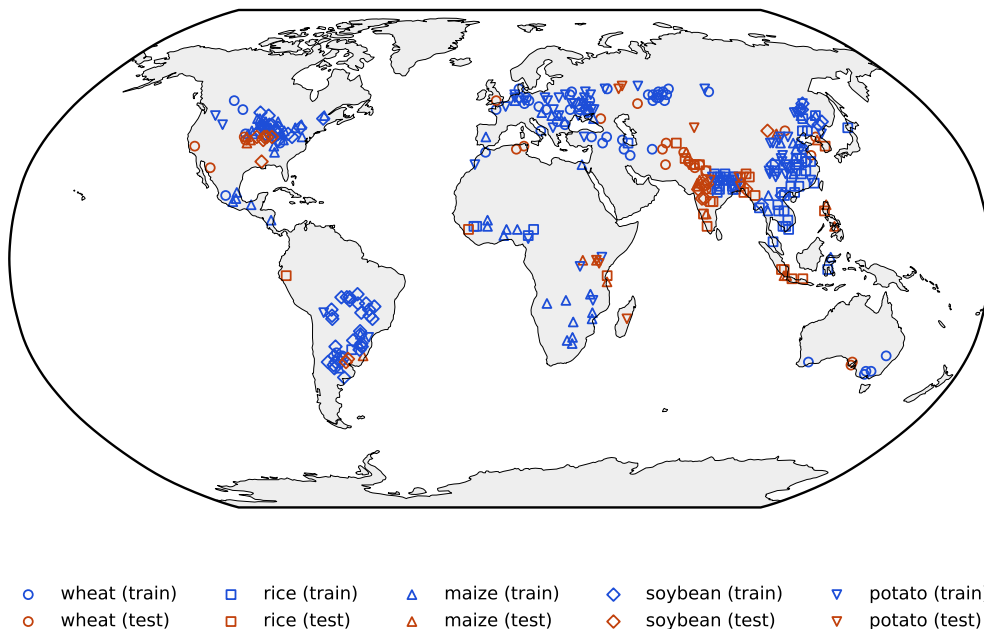


Figure 2: The 400 crop–site pairs, sampled in proportion to the harvested area of each crop (SPAM 2010). Blue: training sites; orange: geographically held-out test sites of the main split.

design) raises the season-level compliance to 90–94% for the light-sharing controllers. It costs about 0.065 in LER for SALS (1.205), which remains the best controller (phenology rule 1.192; Section 3.2).

3.2. Robustness

Seeds and splits. Three training seeds of SALS give LERs of 1.270, 1.268 and 1.269 under the baseline climate and 1.297, 1.298 and 1.297 at +3 °C. On two further spatial splits, in which different blocks of sites are held out (training seed 0; the pooled analysis of Table 3 uses three seeds per split), SALS exceeds the phenology rule by 0.047 and 0.041 in LER under the baseline climate and by 0.052 and 0.039 at +3 °C (main split, season-pooled: 0.034 and 0.032; the site-level paired contrasts of Table S3 are 0.038 and 0.034). In every split SALS ranks first, the phenology rule second and static arrays last.

Canopy cooling. The canopy-cooling coefficient κ is the most uncertain microclimate parameter. We repeated the whole protocol, includ-

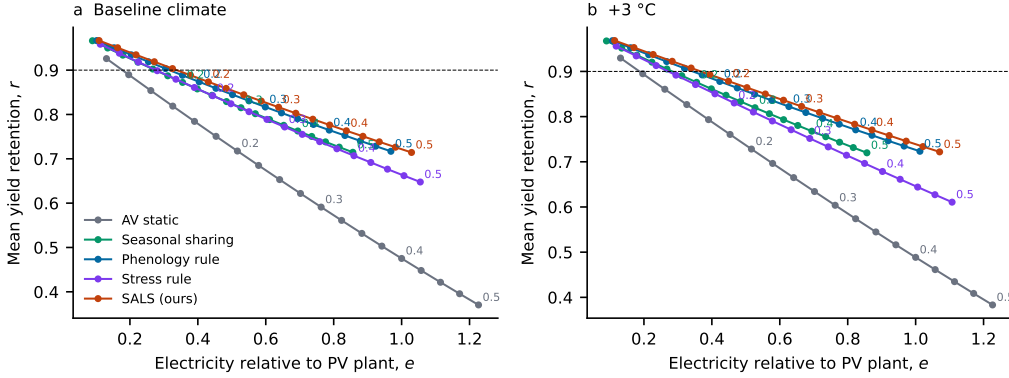


Figure 3: Electricity-retention frontiers at identical array densities (labels: GCR) for the held-out sites. Each point is the mean over test seasons at a fixed GCR; the dashed line is the 90% floor.

ing the historical design step, with $\kappa = 0, 1$ and 2°C . Electricity changes with κ because cooler canopies tolerate denser arrays (SALS: 475, 510 and 549 $\text{MWh ha}^{-1} \text{yr}^{-1}$ under the baseline climate), but the retention is held at 0.905 by the design rule and the ranking is unchanged. SALS exceeds the phenology rule by 0.029, 0.034 and 0.042 in LER under the baseline climate (0.022, 0.032 and 0.051 at $+3^\circ\text{C}$).

Which ingredients matter. Two ablations test the design of SALS at the same training budget. Masking the stress inputs (soil water, ARID, and the maximum temperature, irradiance and ET_0 forecasts) leaves the LER unchanged (1.269 vs 1.270 under the baseline climate, 1.297 vs 1.297 at $+3^\circ\text{C}$). Training only on the baseline climate gives 1.268 and 1.291. The improvement over the rules therefore does not depend on stress-aware inputs or on training in warmed climates.

Further sensitivity analyses (floor conditioning, chance-constrained training, forecast errors, layout, rotation limit, harvest-index assumptions and a floor of 66%) are reported in the Supplementary Material (Section S1); none changes the ranking of the methods.

Tuned rules. The phenology rule above was written, not tuned, whereas SALS is trained. We therefore tuned its window on the training-block seasons with the same objective as the other rules (electricity at the density where mean retention equals the floor), first on a coarse 4×3 grid and then, as a derivative-free baseline, by CMA-ES over a five-parameter schedule (start, end, two ramp widths and a level; three runs of 200 evaluations, Methods).

609 The grid-tuned window starts at 0.15 instead of 0.05 of the thermal-time re-
 610 quirement (Table 3). Two of the three CMA-ES runs, one started at the grid-
 611 tuned window and one at a random point, reached the same almost binary
 612 window (all three runs stopped at the generation cap) from 0.20 to 0.93 of the
 613 requirement at level 1.00; the third stopped in a worse optimum. Pooled over
 614 three splits and 76 distinct crop–site pairs, the grid-tuned window raises LER
 615 over the written rule by 0.026 (95% cluster-bootstrap interval 0.016 to 0.037),
 616 the CMA-ES window by 0.036 (0.023 to 0.052) and SALS by 0.044 (0.030 to
 617 0.060), so the optimisation of a window accounts for about 83% of the margin
 618 of SALS over the written rule. SALS exceeds the CMA-ES window by 0.007
 619 LER (0.002 to 0.014), which is 2.4% more electricity at equal retention, and
 620 the grid-tuned window by 0.018 (0.011 to 0.025). The remaining margin is
 621 small (it covers the pairs only: SALS is the mean of three training seeds
 622 per split, whose pooled margins over the CMA-ES window are 0.007, 0.008
 623 and 0.007 (95% intervals 0.001 to 0.015; results/seed_margins.csv), and the
 624 CMA-ES schedule and the grid window were re-tuned on each split’s own
 625 training block); it is not explained by the tuning of a window or a ramp,
 626 and it cannot come from soil water, forecasts or aridity, because masking
 627 them leaves SALS unchanged (Section 3.2); canopy development and density
 628 remain untested. SALS and the CMA-ES window meet the floor in 83% of
 629 seasons, the written, grid-tuned and seasonal rules in 84–85%; these shares
 630 exceed the 69–76% of Table 2 because the coarser density grid of this analysis
 631 (0.05 against 0.025) selects lower densities, with a mean retention of 0.916
 632 instead of 0.905. Per split, SALS exceeds the CMA-ES window by 0.008
 633 (95% interval 0.001 to 0.016; 32 pairs), 0.009 (0.001 to 0.022; 26 pairs) and
 634 0.005 (−0.002 to 0.012; 35 pairs), with per-pair medians of 0.001 to 0.002,
 635 so the margin is positive on every split but concentrated in a few pairs. The
 636 design step quantises the density to a grid of 0.05 (about 30% of the mean
 637 density), which is much coarser than the margin. We therefore also compared
 638 the controllers on the test seasons at matched mean retention, reading the
 639 LER at a retention of 0.90 off the retention–LER curve of each pair (inter-
 640 polated over the grid): SALS exceeds the CMA-ES window by 0.0033 (95%
 641 interval −0.0004 to 0.0071; 73 pairs, those for which the retention curves of
 642 all compared methods bracket 0.90, instead of the 76 distinct pairs of the
 643 design-based comparison), the grid-tuned window by 0.011 (0.007 to 0.016)
 644 and the written rule by 0.036 (0.023 to 0.051). At fixed densities of 0.10 to
 645 0.30 the difference to the CMA-ES window is 0.001 to 0.003 LER, with inter-
 646 vals that include zero. At matched retention the margin over the optimised

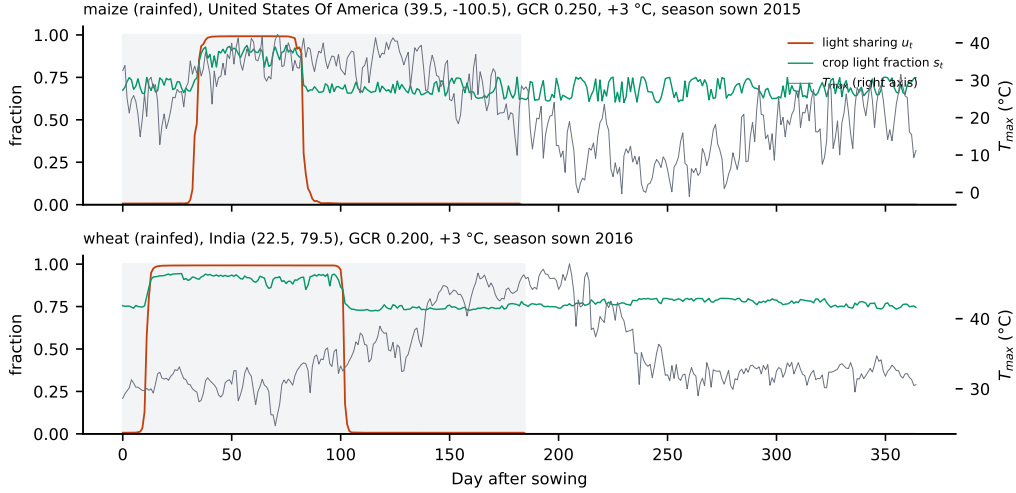


Figure 4: Learned daily control for two test seasons under $+3^{\circ}\text{C}$, at the density chosen for the site, simulated for the dominant sub-field of the site. Shading marks the cropping window. Orange: light-sharing level u_t ; green: fraction of open-field irradiance reaching the crop; grey: daily maximum temperature (right axis).

647 window is therefore not distinguishable from zero.

648 3.3. What the learned policy does

649 Averaged over the rainfed test seasons of the main split at a density of
650 0.3, the light-sharing level of SALS is a smooth function of relative thermal
651 time. It is close to zero in the first 10% of the thermal time required for
652 maturity, rises through 0.55 at 10–20%, stays at 0.94–0.99 between 30% and
653 70%, and falls through 0.58 at 80–90% to about zero at maturity. The plateau
654 overlaps the window in which our crop model penalises shading through the
655 harvest index; we did not test whether this alignment is learned or inherited
656 from the initialisation. The ramps at both ends let the modules produce
657 electricity when additional light is worth least to the crop. Fig. 4 shows two
658 example seasons. The phenology rule, which is also the initialisation of the
659 network, shares light during the whole growing period. Fine-tuning through
660 the simulator found that sharing can be reduced at the edges of the season:
661 this pattern describes the average policy at a density of 0.3 (the designed
662 mean is 0.17) and we did not test that it causes the gain; the CMA-ES
663 window is almost binary, so a smooth ramp is not needed to reproduce most
664 of it.

665 The policy also shares less light on days with a forecast heat stress: the
666 mean level is 0.35 on such days against 0.60 (baseline) and 0.51 (+3 °C)
667 on other days, and 58–63% of sites show a lower level on heat days. This
668 cannot be read as a stress response, however. Heat days occur late in the
669 season, when the phenological ramp is already falling, and a policy without
670 any stress input shows a similar contrast (0.37 vs 0.57 in a separate check).
671 The ablation above confirms that stress information does not improve the
672 result.

673 *3.4. Crops and regions*

674 The benefit of light sharing differs between crops (baseline climate, SALS).
675 Soybean and maize tolerate the densest arrays (GCR 0.24 and 0.19; 691 and
676 577 MWh ha⁻¹ yr⁻¹; LER 1.41 and 1.30), potato follows (0.19; 542; 1.31),
677 and rice and wheat are the most restrictive (0.13 and 0.15; 395 and 428; 1.20
678 for both). The stress rule has a lower LER for every crop (1.26 for soybean,
679 1.14–1.21 for the others).

680 The floor is not met evenly. Production-weighted retention is 0.888 for
681 soybean and 0.898 for maize, below 0.9, because high-yielding seasons lose
682 more than low-yielding ones. Regionally, Latin America and Oceania (4 sites)
683 combine the highest density (GCR 0.41) and electricity (1090 MWh ha⁻¹ yr⁻¹)
684 with a production-weighted retention of only 0.862. Africa and the Middle
685 East (8 sites) reach 0.894, and only 63% of their sites keep a mean reten-
686 tion of at least 0.9. South Asia, the region with the most sites (39), reaches
687 0.902 (67% of sites) at a density of 0.14. The regional figures rest on 4 to 39
688 sites and are uncertain: the 95% site-bootstrap intervals of the production-
689 weighted retention are 0.841 to 0.897 for Latin America and Oceania, 0.868
690 to 0.908 for Africa and the Middle East and 0.898 to 0.905 for South Asia,
691 and the share of sites that meet the floor has intervals of 25 to 88% (Africa
692 and the Middle East), 25 to 100% (Latin America and Oceania) and 51 to
693 79% (South Asia; results/by_region_ci.csv). Only the retention deficit of
694 Latin America and Oceania and of Africa and the Middle East relative to
695 South Asia is therefore suggestive, and the region comparison should be read
696 as an indication. A regulatory floor enforced per site would therefore need a
697 region- and crop-specific design step, not a global one.

698 *3.5. Model behaviour and validation*

699 **PV output.** The simulated specific yield of an isolated tracker row
700 agrees with PVGIS 5.2 (European Commission, Joint Research Centre, 2022)

701 at 29 sites with $r = 0.971$, a mean bias of -4.9% and a mean absolute
 702 deviation of 5.5% (Section 2.2). At higher ground coverage the shading and
 703 diffuse-transposition model was compared with pvlib (Holmgren et al., 2018)
 704 (infinite-sheds model with isotropic sky and backtracking, driven by the same
 705 hourly irradiance and sun position at 12 sites spanning the latitudes of the
 706 data set, year-round). The simulated plane-of-array irradiance is 0.4% above
 707 pvlib at $g = 0.05$ and 0.7 , 1.4 , 2.1 , 2.6 and 3.0% above it at $g = 0.1$, 0.2 ,
 708 0.3 , 0.4 and 0.5 (site ranges within 0.1 to 1.0 points of these means). The
 709 loss relative to a sparse array is therefore underestimated: 2.8% instead of
 710 4.4% at $g = 0.3$ and 6.3% instead of 8.7% at $g = 0.5$, which favours dense
 711 arrays by about 1.5 to 2.5 points of electricity in this range. To bound the
 712 effect of this bias we multiplied the electricity of every method by the ratio
 713 of the pvlib to the simulated relative irradiance (0.996 at $g = 0.1$, 0.990
 714 at 0.2 , 0.983 at 0.3 , 0.978 at 0.4 and 0.974 at 0.5 ; retention and hence the
 715 designed densities do not depend on electricity), and, as a bound that lets
 716 the shading loss of DC energy exceed that of irradiance, by the square of
 717 this ratio. The LER of the light-sharing methods falls by 0.003 to 0.004
 718 (0.005 to 0.007 with the bound; SALS 1.251 to 1.247 , 1.244), that of static
 719 arrays by 0.0002 , and the margins of SALS over the written rule and the
 720 CMA-ES window are unchanged within 0.001 (0.043 and 0.007). Break-even
 721 prices rise by 0.8 USD MWh $^{-1}$ (1.5 with the bound; SALS 73.4 , 74.2 ; static
 722 arrays 58.6 , 58.7), the net value of SALS at 90 USD MWh $^{-1}$ falls from 7.6 to
 723 7.2 (6.8) kUSD ha $^{-1}$ yr $^{-1}$, and the ranking of the methods does not change
 724 (results/pvbias.csv).

725 **Shade response.** Fig. 5 shows the simulated relative yield of rainfed
 726 crops under static arrays as a function of the season-mean shading of the crop.
 727 Yield declines close to proportionally with shading: $84\text{--}89\%$ relative yield at
 728 12% shading, $70\text{--}78\%$ at 25% , $57\text{--}67\%$ at 37% and $35\text{--}46\%$ at 57% . Wheat
 729 loses most and potato and maize least. This response is steeper than the *less*
 730 *than proportional* losses that Laub et al. (2022) report for small shade levels
 731 of C3 cereals and tubers, and it does not reproduce their finding that maize
 732 and grain legumes are more shade sensitive than C3 cereals. In the simulator,
 733 shading also relieves heat and drought in rainfed maize and soybean, which
 734 offsets the higher harvest-index penalty we assigned to them. The absolute
 735 effect of shading on each crop is therefore uncertain (Section 4.5).

736 **Comparison with field trials.** We fitted one radiation-use compensa-
 737 tion coefficient per crop to the yield–shading curves of Laub et al. (2022) using
 738 training-block sites only (Methods, calibration script in the repository). The

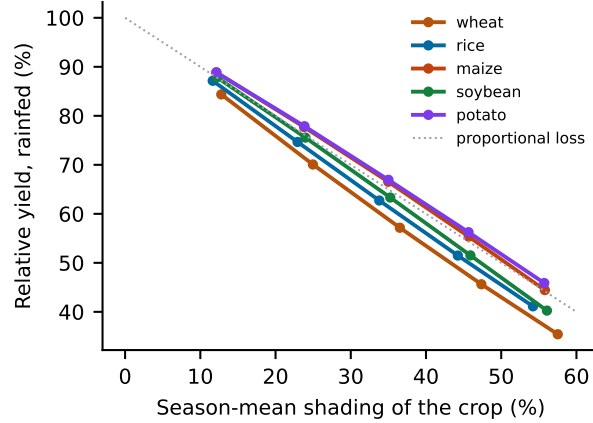


Figure 5: Simulated relative yield of rainfed crops under static arrays (GCR 0.1–0.5) against season-mean shading of the crop, baseline climate, held-out sites. The dotted line is proportional loss.

fit changed wheat and rice only (coefficients 0.4 and 0.1) because the proportional model already lies within about 0.02 of the C3-cereal and tuber curves, and it cannot lower the maize and legume responses below the proportional case (root-mean-square error 0.16 for maize and 0.08 for soybean). Against field trials (not independent of the choice of ensemble, Table 4) (Table 4) the model underestimates retained yield: rice 0.79 against 0.84 (Gonocruz et al., 2021), wheat and potato 0.74 and 0.72 against 0.81–0.82 in the worse season of Weselek et al. (2021) (and above 1 in the better one), and maize 0.70–0.75 against 0.83–0.91 at 30–35% shading (Ramos-Fuentes et al., 2023), where harvest index was not affected by shade. The model is therefore conservative for food retention with respect to these trials, except for the stronger maize losses of the meta-analysis. Table S6 shows the consequence for the design results: with the calibrated response the allowed density and the LER of every method rise (phenology rule 1.28 against 1.23), and with a response fitted to the field trials the LER of the phenology rule is 1.62. The phenology rule is the best method in all three cases, and light sharing adds LER +0.17, +0.19 and +0.30 over static arrays (conservative, calibrated, field-fitted); only the order of the seasonal and stress rules changes. The absolute gains for rice, wheat and potato are thus conservative with respect to these trials; for maize and soybean the meta-analysis points the other way (Section 3.6).

3.6. *A shade-response model learned from field data*

Cross-validation by study shows what the data can and cannot support (Tables S4 and S5). Over all 339 shaded observations the root-mean-square error of a held-out study falls from 0.43 for proportional loss $y = 1 - s$ to 0.37 for one curve for all crops and to 0.31 for crop-type curves, but this gain comes from crop types that the simulator does not represent (berries, forages, fruits and vegetables: 0.53 to 0.36). For the four crop types that the simulator uses, the learned curves are not better than proportional loss: the error is 0.222 against 0.220 pooled (difference +0.002, 95% interval -0.023 to $+0.030$), 0.187 against 0.214 for C3 cereals, and slightly worse for maize, grain legumes and tubers. Partial pooling is as good as separate curves (0.312 against 0.313) and the covariates add almost nothing (0.311). Gradient-boosted trees on the same inputs are worse (0.321) than the parametric model. The error of 0.22 for the four crop types is large: yields of a new study differ from the learned curve by about 20 percentage points, which is the between-study spread that a design under uncertainty has to cover. Crop type is the information that transfers between studies, and none of the covariate effects on $\log \beta$ is clearly different from zero (irradiation -0.10 , 90% interval -0.32 to 0.09 ; subtropical zone -0.37 , -0.73 to 0.07 ; intercropping 0.16 , -0.48 to 0.45). The estimated effect of solar-panel shade is not interpretable, because all 46 panel observations are of leafy vegetables, mainly lettuce, and the effect is confounded with crop type.

The learned curves show a clear ordering of crop types (Table 5). At 20% shading C3 cereals and tubers lose less than proportionally (relative yield 0.93), whereas maize (0.73) and grain legumes (0.71) lose more; at 40% shading the values are 0.72 and 0.71 against 0.42 and 0.40. Intervals are wide for maize (0.27–0.59 at 40%) and for tubers, which rest on two studies. The simulator’s responses are close to the data for cereals and potato but, under both calibrated ensembles, too mild for maize at 40% shading (0.75 for E2 against 0.42 in the data); for soybean the E2 value (0.48) lies within the interval of the data (0.40, 80% interval 0.34–0.53). The calibration cannot lower the maize response further, because the simulated shading also relieves heat and drought stress and the compensation and harvest-index parameters are bounded below at zero. The field trials of maize are also milder than the meta-analysis: Ramos-Fuentes et al. (2023) report a harvest index unaffected by shade, and Sekiyama and Nagashima (2019) a yield 5.6% above the control under a low-density stilt-mounted array in one season at one site. For maize and soybean the direction of the bias is uncertain, and the gains reported in

797 this paper are possibly overestimated.

798 3.7. Learning under shade-response uncertainty

799 **Calibration.** Gradient descent through the simulator fitted the five crops
800 to the data-driven curves and the field trials (ensemble E2, Table S7). The
801 data identify two features: the harvest index of maize is hardly affected by
802 shade (multiplier 0.16 ± 0.13 , $c \approx 0.14$), as observed by Ramos-Fuentes et al.
803 (2023), and soybean is more shade sensitive than the cereals (compensation
804 0.05, harvest-index multiplier 1.66), as in Laub et al. (2022). Wheat has the
805 largest compensation (0.91 ± 0.31) and the largest uncertainty; the compen-
806 sation is an effective parameter that absorbs diffuse-light and light-saturation
807 effects that the simulator does not model explicitly, not a physical value.

808 **Learned controllers.** Controllers were evaluated under four fixed re-
809 sponses on 21 crop–site pairs of split 0 with one training seed, so the intervals
810 below, which are cluster bootstraps over pairs of per-pair mean LER differ-
811 ences, do not include seed or split variation (Table 6 gives season means,
812 which differ slightly). Under every response SALS, trained on the propor-
813 tional response, is better than the written phenology rule: by +0.019 (+0.007
814 to +0.039), +0.024 (+0.009 to +0.046), +0.038 (+0.018 to +0.062) and
815 +0.015 (+0.001 to +0.037) LER under the proportional, calibrated, field-
816 fitted and harsh responses. It is also above the grid-tuned window and the
817 CMA-ES optimised window, but by less: 0.005 to 0.018 and 0.005 to 0.010
818 LER, with intervals that exclude zero for the grid-tuned window under 3 of
819 the four responses and for the CMA-ES window under none. Training on
820 the narrow ensemble E1 did not change this: SALS-robust differs from SALS
821 by -0.005 to $+0.003$ (intervals contain zero except for one response) and
822 SALS-adaptive, which sees the measured biomass ratio, by -0.010 to $+0.007$.
823 Training on the wider ensembles did not help either (the robust controllers
824 fall below SALS even under responses inside their training ensemble, which
825 may point to under-training with our budget of 300 steps): SALS-robust
826 on E2 and E3 and SALS-adaptive on E3 are below SALS by 0.014, 0.017,
827 0.017 LER under the proportional response, 0.017, 0.021, 0.020 under the
828 calibrated response, 0.004, 0.005, 0.011 under the field-fitted response (in-
829 tervals contain zero) and 0.008, 0.014, 0.017 under the harsh response; the
830 intervals exclude zero in 7 of the 9 comparisons for the proportional, cali-
831 brated and harsh responses. The E3 robust controller is not distinguishable
832 from the written rule ($+0.001$ to $+0.032$) under any response (all intervals
833 contain zero). With one seed and 21 pairs these controllers therefore did not

834 retain the margin of SALS within our training budget; a controller that must
835 perform across plausible responses is less specialised, but a larger budget or
836 more seeds could change this. The feedback rule is worse than the written
837 rule by 0.026–0.058 LER under every response (intervals exclude zero). A
838 plausible reason is that the biomass ratio does not reveal the harvest-index
839 loss from shading around flowering, but we did not test this explanation.

840 **Design under uncertainty.** The design step is a second source of risk,
841 and compliance there is sensitive to the assumed response (Tables 7 and 8,
842 Fig. 6). The designs below are evaluated by leaving one ensemble member
843 out as the truth and using the other four, so that the truth never enters the
844 design. Pooled over the three spatial splits (76 distinct crop–site pairs), a
845 density chosen under the mean response of E2 leaves 70% of the pairs at the
846 floor when the true response is another plausible one, against 87% for an
847 oracle design (E3: 71% and 85%). Choosing the density so that the floor is
848 met in at least 75% of the other members raises compliance by +7.9 points
849 under E2 (95% cluster-bootstrap interval 4.5 to 11.6) and +7.9 under E3
850 (3.9 to 12.1), at a cost of 0.031 and 0.025 in LER. Raising the floor used
851 in the design from 0.90 to 0.92 under the mean response does more: +18.0
852 points (13.4 to 22.6) at 0.068 LER under E2 and +14.3 points (9.5 to 19.2) at
853 0.056 under E3, and the shortfall is smaller (mean over pairs of the maximum
854 of zero and 0.9 minus retention: 0.0022 against 0.0045 for the quantile rule
855 under E2). The two kinds of rule lie on one frontier (Fig. 6). Interpolating the
856 margin frontier (floors 0.90 to 0.96) at the compliance of the ensemble rules,
857 the LER of the margin minus that of the rule is +0.001 (-0.009 to +0.012) for
858 the rule with 75% of the other members and +0.003 (-0.016 to +0.021) for the
859 rule with all of them under E2, and -0.006 (-0.019 to +0.005) and +0.024 (-
860 0.011 to +0.058) under E3 (95% intervals). We find no detectable difference,
861 but the intervals (down to -0.02 and up to +0.06 LER) are too wide to
862 establish equivalence. The ensemble is thus useful for sizing the margin (0.02
863 to 0.04 above the floor for 85–97% of the pairs), not for replacing it. None
864 of these rules is safe outside the ensemble (Table 8): under the proportional,
865 calibrated and harsh responses the mean-response design leaves 7–17% of the
866 pairs at the floor, a floor of 0.92 or the ensemble rules 12–33%, a floor of 0.94
867 29–47% and a floor of 0.96 51–78%, at a cost of about 0.2 LER, whereas the
868 field-fitted response, which lies within the range spanned by the ensembles,
869 is handled well by every rule (80–99%) except the mean-response design (68–
870 79%). Outside the ensemble the shortfall of the pairs below the floor is also
871 large: under E2 the mean shortfall per pair (maximum of zero and 0.9 minus

872 retention) is 0.075, 0.060 and 0.104 for the mean-response design under the
 873 proportional, calibrated and harsh responses, against 0.053, 0.038 and 0.078
 874 for a floor of 0.92 and 0.012, 0.007 and 0.024 for a floor of 0.96. Across the
 875 five members of an ensemble, the standard deviation of the simulated relative
 876 yield of a crop (static arrays, baseline climate, held-out sites of split 0) is 0.02
 877 to 0.04 at 20% shading and 0.02 to 0.06 at 40% under E2, and 0.02 to 0.09
 878 and 0.03 to 0.15 under E3 (largest for soybean and potato; ranges between
 879 the extreme members up to 0.36), against a leave-one-study-out error of 0.22
 880 for new studies. The two quantities differ (a spread of curves against an error
 881 over observations), but the ensembles are narrower than the between-study
 882 spread for all crops except, possibly, soybean under E3. The pooled shares
 883 hide large differences between crops (Table S9, E2, 10 to 20 pairs per crop).
 884 Under the proportional, calibrated and harsh responses the design under
 885 the mean response and the rule that requires 75% of the members leave no
 886 wheat, rice or potato pair at the floor and at most 12% of the maize pairs (a
 887 floor of 0.92 raises this to at most 10% of the rice pairs and 18% of the maize
 888 pairs), whereas 90 to 100% of the soybean pairs meet it, because the ensemble
 889 already contains steep soybean responses (the oracle design reaches 90% for
 890 soybean and 65 to 100% for the other crops). Even a floor of 0.96 leaves
 891 0 to 12% of the wheat pairs at the floor under the harsh and proportional
 892 responses and 42% of the potato pairs. The out-of-ensemble failure is thus
 893 a failure for cereals, tubers and maize, and the pooled values of 7 to 33%
 894 mix crops that fail completely with a crop that does not fail. The guarantee
 895 therefore holds only within the range of responses that the ensemble spans;
 896 we did not test an ensemble that spans the proportional and harsh responses,
 897 which would protect against them at a higher cost in LER.

898 *3.8. Economics of light sharing*

899 All figures in this section follow from the assumed parameters of Table 9
 900 and should be read as a sensitivity analysis, not a forecast. The economics
 901 are dominated by the capacity installed, not by the crop (Table 10): the
 902 annuity and operating cost of the modules is 11–32 kUSD per hectare and
 903 year, whereas the crop revenue lost is about 0.3 kUSD, so a method is worth
 904 what it earns per installed watt. With the densities designed for the floor,
 905 static arrays earn +0.3 kUSD ha⁻¹ yr⁻¹ at 60 USD MWh⁻¹ and break even
 906 at 59, whereas all light-sharing methods lose money at that price (−3 to
 907 −10 kUSD) and break even at 70 (stress rule), 73 (SALS), 73 (CMA-ES
 908 window), 77 (grid-tuned window), 80 (written rule) and 92 (seasonal sharing)

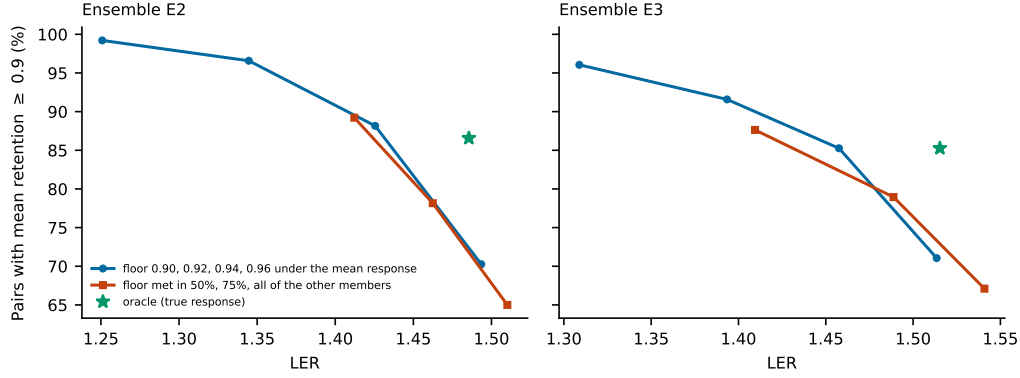


Figure 6: Share of crop-site pairs at the floor against LER for the phenology rule under design rules for shade-response uncertainty (leave-one-member-out, three splits, 76 pairs): a higher floor under the mean response (blue; the first point is the design under the mean response) and a floor met in a share of the other members (orange), with the oracle design (green).

909 USD MWh⁻¹. The 95% cluster-bootstrap intervals over pairs are 56–61 for
 910 static arrays, 69–76 for SALS, 70–77 for the CMA-ES window and 75–84 for
 911 the written rule, and 46–50 for the plant (break-even prices of SALS and the
 912 CMA-ES window differ by -0.4 , -1.1 to 0.4); they cover the sampling of
 913 pairs and not the cost assumptions. At 90 USD MWh⁻¹ static arrays (6.3
 914 kUSD) are better than the written (4.0) and grid-tuned (5.5) rules, close
 915 to the CMA-ES window (7.3) and worse than SALS (7.6), which overtakes
 916 static arrays above about 85 USD MWh⁻¹; the difference between SALS and
 917 static arrays at 90 USD MWh⁻¹ is $+1.3$ kUSD (95% interval -0.04 to 2.7 , 76
 918 pairs) and not distinguishable from zero, whereas SALS exceeds the CMA-
 919 ES window by 0.3 (-0.1 to 0.8 , not distinguishable from zero), the grid-
 920 tuned window by 2.1 (1.6 to 2.7) and the written rule by 3.6 (2.8 to 4.5). A
 921 conventional plant that replaces the crop breaks even at 48 USD MWh⁻¹ and
 922 is more profitable than SALS above about 35 USD MWh⁻¹ and than static
 923 arrays above about 46 (16 kUSD at 60, 94 kUSD at 120, against -6 and 21
 924 for SALS); below 48 the plant loses money, and below about 46 static arrays
 925 lose least of the options (at 30 USD MWh⁻¹: plant -23.2 , SALS -18.7 ,
 926 static -5.7 kUSD). These plant results rest on the assumed capex of 0.75
 927 USD Wp⁻¹ for a ground-mounted tracker plant against 1.0 for the elevated
 928 agrivoltaic structure; the higher agrivoltaic capex is in line with a levelised
 929 cost of electricity that is 38% higher than that of ground-mounted PV in a

930 techno-economic study (Schindele et al., 2020); with equal capex the plant
931 breaks even at about 63 USD MWh⁻¹ and overtakes SALS above about 58
932 and static arrays above about 64. Per megawatt-hour, the electricity of SALS
933 costs about 73 USD including the crop lost, against 48 USD for the plant, a
934 difference of 25 USD MWh⁻¹. A premium of that size would equalise the unit
935 cost but not the profit per hectare, because a hectare of plant produces three
936 times as much electricity (1306 against 438 MWh): for the same electricity
937 as one hectare of plant, SALS needs about three hectares of array, which
938 costs about 25 USD MWh⁻¹ more including the crop lost but keeps about
939 90% of the yield of that cropland; relative to the plant, SALS would need a
940 premium of about 49 USD MWh⁻¹ at a market price of 60 and about 109 at
941 90 to match its profit per hectare, and only about 13 at 60 (none above 73) to
942 match the open field. The sensitivity is large (Table S8), and the ranking of
943 static arrays and light sharing flips with the module cost (at 1.5 USD Wp⁻¹
944 static arrays are better than SALS, at 0.7 SALS leads by 7 kUSD): at 90
945 USD MWh⁻¹ the net value of SALS changes from +17 to -8 kUSD between
946 module costs of 0.7 and 1.5 USD Wp⁻¹, whereas crop prices matter little.

947 Because the density was designed for the floor and not for profit, we also
948 let each site choose, among the densities that meet the floor in its history, the
949 one that maximises its historical net value (sites where none pays stay open
950 field; Fig. 7). Then no method pays at 30–40 USD MWh⁻¹; at 60 14–59% of
951 the pairs install an array (none for seasonal sharing) and the net values are
952 small (static arrays 0.9, SALS 0.3 kUSD); static arrays are the best option up
953 to about 75 USD MWh⁻¹, SALS overtakes them at about 75 (4.9 against 4.3
954 at 80, 8.3 against 6.3 at 90, 20.7 against 12.3 at 120), the CMA-ES window
955 at about 78, the stress rule at about 81, the grid-tuned window at about
956 87 and the written rule at about 97. The value of control is therefore real
957 but depends on a high price: at 90 USD MWh⁻¹ and with profit-optimal
958 densities SALS exceeds the CMA-ES window by 0.4 kUSD ha⁻¹ yr⁻¹ (95%
959 cluster-bootstrap interval 0.01 to 0.8), the grid-tuned window by 1.7 (1.2 to
960 2.2), the written rule by 2.8 (2.0 to 3.6) and static arrays by 2.0 (0.8 to 3.3);
961 the CMA-ES window exceeds static arrays by about 1.6 (0.4 to 2.8), so most
962 of the value of control is obtained with a well-optimised rule.

963 **Cost of the safety margin.** At 90 USD MWh⁻¹, moving from a density
964 designed under the mean response to a floor of 0.92 under that response costs
965 1.00 kUSD ha⁻¹ yr⁻¹ (17%) under E2 for 18 more points of compliance (70%
966 to 88%) and 1.13 kUSD (20%) under E3 for 14 points (71% to 85%), about
967 56 and 79 USD ha⁻¹ yr⁻¹ per point. The ensemble rule that requires all other

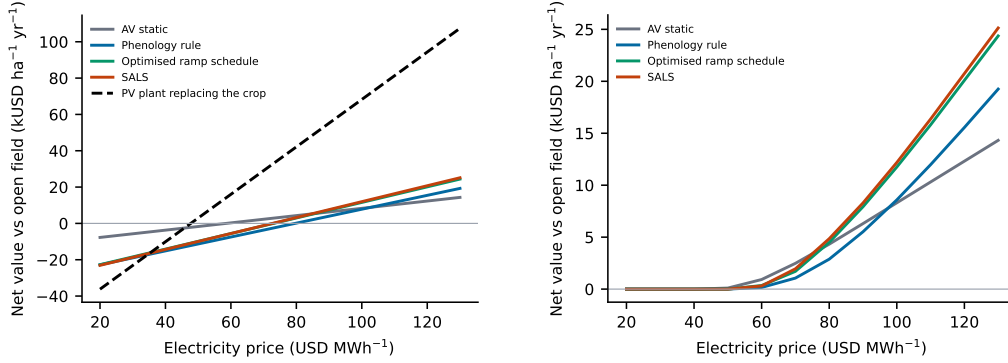


Figure 7: Net value per hectare relative to the open field against electricity price. Left: densities designed for the floor, with a conventional plant replacing the crop (dashed). Right: each site chooses the feasible density that maximises its historical net value. Illustrative parameters of Table 9.

members to meet the floor costs 0.85 and 1.22 kUSD for 19 and 17 points (45 and 74 USD per point), so on this measure it is cheaper than the margin under E2 and similar under E3. At 60 USD MWh⁻¹ a margin has a negative cost, because every module beyond break-even loses money; the economics of the design margin therefore depend on the price level, whereas its effect on compliance does not.

3.9. Agrivoltaics under warming

Under warming, the open-field production of the held-out sites (Fig. 8) falls at +3 °C by 23% for maize and 18% for rice, is nearly unchanged for soybean, and rises for wheat (+3%) and potato (+10%). Agrivoltaic production with light-sharing control follows the open field at about 90% of its level, so it inherits the same climate impacts (maize -31%, rice -27% relative to the baseline open field at +3 °C). Agrivoltaics does not protect staple production against warming in this model. The LER of light-sharing systems increases slightly with warming (SALS 1.270 to 1.297; phenology rule 1.236 to 1.265), because the floor allows slightly denser arrays in warmer conditions (GCR 0.173 to 0.177 and electricity 510 to 540 MWh ha⁻¹ yr⁻¹). The margin of SALS over the rules does not grow with warming (Table S3).

3.10. Global potential and what it would take

Every site is evaluated when it is held out (222 of the 400 crop-site pairs are held out in at least one split, and three of them have no valid season,

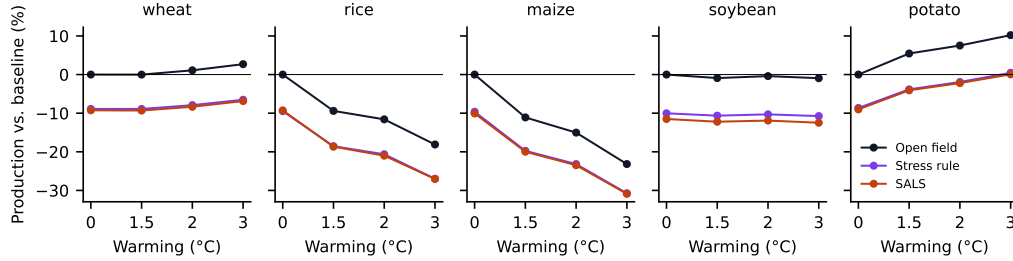


Figure 8: Production relative to the baseline open field (sum over held-out site-seasons) as a function of warming, for the open field, the stress rule and SALS.

leaving 219), and each crop’s mean electricity per hectare is scaled to its physical area in SPAM 2010 (581 Mha for the five crops, which avoids counting multiple-cropped land twice). Equipping 5% of this area (29.1 Mha) with SALS arrays at the floor would produce 14.4 PWh yr^{-1} under the baseline climate and 16.0 PWh yr^{-1} at $+3^\circ\text{C}$, with a production-weighted retention of 0.903 and 0.900. The phenology rule would give 13.0 and 14.5 PWh yr^{-1} , and static arrays 6.4 and 6.9 PWh yr^{-1} . For comparison, world electricity generation was about 30 PWh in 2023 (Ember, 2024).

These numbers require large amounts of hardware. SALS at 5% coverage needs 10.6 TWp of modules (11.3 TWp at $+3^\circ\text{C}$), at a capacity factor of 15.5% (16.2%). World solar PV capacity at the end of 2024 was about 1.9 TW (IRENA, 2025), so the requirement is about 5.6 times the installed PV fleet. The electricity is also variable, peaks at midday and would be curtailed or stored. The figures are technical potentials, not projections. Ordering sites by electricity per hectare gives a supply curve (Fig. 9): the best 2% of the staple cropland would supply about 21 PWh yr^{-1} and the best 4.7% about 40 PWh yr^{-1} with SALS. The best tenth of the sites are mostly rainfed maize and soybean sites in the Americas and in southern and eastern Africa (7% mean irrigated fraction against 29% overall), which are also among the regions where the floor is hardest to meet (Section 3.4). Siting by electricity alone would therefore concentrate agrivoltaics where food-security risk is highest.

Maize and soybean, for which the simulated shade response is milder than the meta-analysis (Section 3.6), account for 53% of the electricity of the upscaling (a scaling convention of 5% of the cropland, reported without an interval) (maize 30%, soybean 23%, wheat 29%, rice 15%, potato 3%).

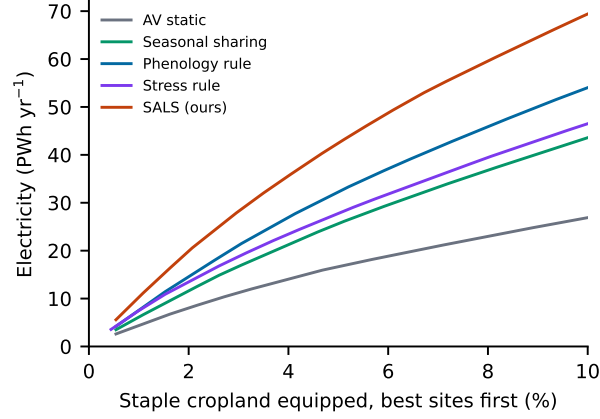


Figure 9: Supply curve under the baseline climate: cumulative electricity when sites are equipped in order of decreasing electricity per hectare, against the share of staple cropland (physical area) equipped. Each controller is evaluated at the density that keeps its mean retention at 0.9.

Counting only wheat, rice and potato would give about 7 PWh yr⁻¹ at the same coverage, so the global figure should be read as an upper estimate for about half of its content. By crop, the LER of SALS under the baseline climate is 1.20 for wheat and rice, 1.31 for potato, 1.30 for maize and 1.41 for soybean.

4. Discussion

4.1. What the results establish

Four findings are discussed; the first three stand on consistent evidence and the fourth is not established. First, sharing light with the crop is the main lever on the electricity that tracker-based agrivoltaic arrays can produce under a yield floor. With static trackers the floor limits arrays to about 6% ground coverage and an LER of 1.07. Any light-sharing control allows 2.1 to 2.9 times the density, and the LER rises to 1.18–1.24 for hand-written rules. Second, the best simple rule follows crop phenology. It shares light from the onset of canopy development (5% of the thermal-time requirement) until maturity and produces electricity before and after, when the crop does not use the light. It needs only the crop's thermal time, which is observable on the farm, and it is insensitive to forecast errors, whereas the rule that reacts to heat and drought stress is worse in every climate and degrades with

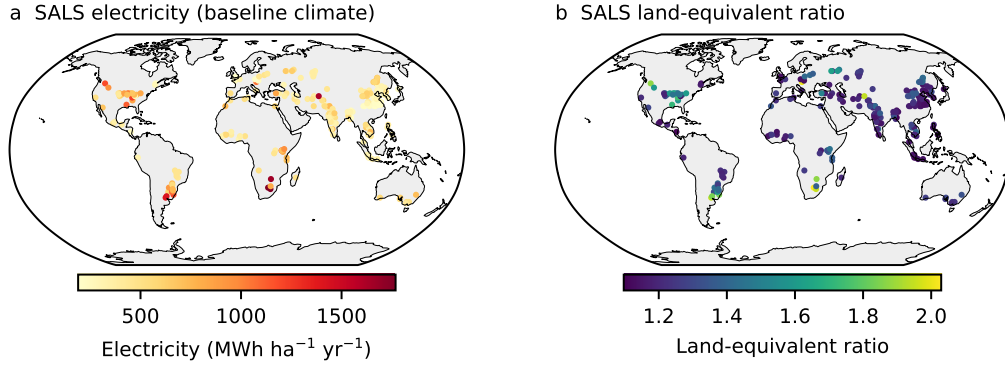


Figure 10: SALS electricity (a) and land-equivalent ratio (b) at the pooled held-out sites (baseline climate, mean of seasons 2015–2019).

forecast error. Third, the qualitative result does not depend on the weakest part of the model. The simulated shade response is conservative against the field trials we could use (Section 3.5); calibrating it raises the allowed density, and the phenology rule stays the best method, with a gain over static arrays of 0.17, 0.19 and 0.30 LER under the conservative, calibrated and field-fitted responses. However, the response learned from the meta-analysis shows steeper losses for maize and grain legumes than the simulator reproduces (Section 3.6), so the absolute gains are conservative for wheat, rice and potato and uncertain, possibly overestimated, for maize and soybean (53% of the upscaled electricity). Fourth, a controller learned through the differentiable simulator is better than the best rule we could write, by 0.044 in LER against the written rule but by only 0.007 (0.002 to 0.014; 2.4% more electricity at equal retention, and not distinguishable from zero at matched retention) against a window optimised on the training seasons of each split. The margin over the written rule is consistent over training seeds (main split, range 0.002 in LER), the three spatial splits, three canopy-cooling assumptions and the baseline and $+3^\circ\text{C}$ climates, whereas the margin over the optimised window is small and varies between splits (0.008, 0.009 and 0.005 LER on splits 0, 1 and 2, mean of three training seeds; per-pair medians 0.001 to 0.002), so it is not established as a reproducible gain. This gain is small compared with the gap between static and dynamic control.

1055 4.2. *What role does artificial intelligence play?*

1056 We tested three ways in which learning could matter, and the answers dif-
1057 fer. The first is learning the crop’s response to shade from data. The learned
1058 model is useful mainly as a measurement of how little can be learned: a hier-
1059 archical model fitted to 428 observations from 58 studies predicts a held-out
1060 study of the four staple crop types with an error of 0.22 in relative yield, no
1061 better than proportional loss, and the gain over proportional loss (0.43 to
1062 0.31 over all types) comes from crop types that the simulator does not repre-
1063 sent. Crop type is the information that transfers, whereas irradiation, climate
1064 zone and shade type add nothing measurable and a non-parametric model
1065 (gradient-boosted trees) is worse with this number of studies (Section 3.6).
1066 Two things follow. The shade response of the staple crops is a distribution
1067 with a spread of about 20 percentage points between studies and not a curve,
1068 and the process-based simulator does not reproduce the steep losses of maize
1069 and legumes seen in the data. Learning the response was limited by the data
1070 and not by the model: two studies support the tuber curve, no rice study
1071 exists, and 46 panel observations are all leafy vegetables.

1072 The second is calibration through the differentiable simulator. Gradient
1073 descent through whole seasons fitted the response of each crop to the learned
1074 curves and to the field trials, and it produced an ensemble of plausible re-
1075 sponses that fits the trials and the curves for cereals and potato but not the
1076 maize and soybean curves (Section 3.7). The ensemble informs the design
1077 decision: a density chosen under the mean response leaves about 70% of
1078 crop–site pairs at the floor if another plausible response holds, against 85–
1079 87% for an oracle, and a density chosen to meet the floor in 75% of plausible
1080 responses recovers about half of that gap at a cost of 0.03 LER. A simple
1081 margin reaches similar compliance: raising the floor from 0.90 to 0.92 under
1082 the mean response reaches 85–88% of the pairs, and interpolating the margin
1083 frontier at the compliance of the ensemble rules gives LER differences (mar-
1084 gin minus rule) of +0.001 (−0.009 to 0.012) and +0.003 (−0.016 to 0.021)
1085 under E2 and −0.006 (−0.019 to 0.005) and +0.024 (−0.011 to 0.058) under
1086 E3, so the two are not distinguishable on this sample (equivalence is not
1087 established, since the intervals reach ± 0.02 to 0.06 LER). None of the rules
1088 is safe outside the ensemble (7–33% of the pairs at the floor under the pro-
1089 portional, calibrated and harsh responses for the mean design, the margin of
1090 0.92 and the ensemble rules), and only a floor of 0.96 reaches 51–78% at a
1091 cost of about 0.2 LER. What the ensemble adds is the size of the margin,
1092 not a better rule. The quantile rule is a scenario-type design (Calafiore and

Campi, 2006; Campi and Garatti, 2008); with five members it has none of the finite-sample guarantees of that theory, and since the comparison with the margin shows no detectable difference (with intervals too wide to establish equivalence), we see no evidence that it is needed. The bottleneck of tracker-based agrivoltaics under a yield-retention floor is the uncertainty of the crop response that enters the design, and it is data on that response, not larger networks, that would improve the result; the learned curves show a spread of about 20 percentage points between studies. The guarantee holds only within the range of responses that the ensemble spans.

The third is control, and here learning contributes little. Differentiability is what lets a network of 5441 weights be trained in 300 gradient steps through whole seasons; a derivative-free search such as CMA-ES handled a five-parameter schedule with 200 simulator evaluations per run and found an almost binary window that is only 0.007 LER behind SALS, but we did not test it on the network, whose 5441 weights would need gradient information. The learned controller adds 0.007 LER over an optimised window (and 0.017 over a grid-tuned one, 0.044 over the written rule), and 0.005–0.010 over the optimised window under the four alternative responses on one split (21 pairs, intervals containing zero); a reader who needs a simple, auditable controller can use that window and obtain about 84% of the LER gain of SALS over the written rule. Neither of the two extensions that we expected to matter under uncertainty did: training on ensembles of responses (E1, E2 and E3) did not help and, for the wider ensembles, did not retain the margin over the written rule within our training budget (one seed, 21 pairs), and neither did giving the controller a farm-measurable signal, the biomass of the crop relative to a reference plot, nor a feedback rule built on that signal. The learned policy also has to be initialised from the phenology rule (Section 4.5). We therefore do not claim that learned control is what makes agrivoltaics work: in this model, the schedule that a learned policy reaches is a smooth function of crop phenology that an optimised window already approximates to within 0.007 LER, and the source of the remaining margin is not identified.

4.3. What the results do not establish

The early hypothesis that stress-aware control, that is, shading the crop on heat and drought days, would be a main source of benefit under warming was not supported. The stress rule performs worse than the phenology rule in every climate, and a learned controller without stress inputs performs as well as one with them. A plausible reason is that in our crop model the yield

gain from shading on a hot or dry day is smaller than the loss of light on that day, so stress information does not help. Whether this holds for crop models with more detailed canopy-temperature and heat-stress responses is an open question. Agrivoltaics does not protect staple production against warming in our simulations: light-sharing systems produce about 90% of the open-field yield in every climate and follow the same decline in maize and rice (Fig. 8).

4.4. Food security in practice

Holding the historical mean retention at 90% does not hold it in each season or at each site. 70% of SALS seasons meet the floor, the lower tail is slightly heavier than for the rules, and production-weighted retention is below 0.9 for soybean (0.888) and maize (0.898) and in several regions. Chance-constrained design raises season-level compliance to about 92% at a cost of 0.065 in LER, and the ranking of the controllers is preserved. Whichever formulation is chosen, a per-site regulatory floor would need a region- and crop-specific design step. The upscaled potential (Section 3.10) shows the trade-off in magnitude. The 14 PWh yr⁻¹ that 5% of the staple cropland could supply need about 10 TWp of modules, several times the present global fleet, and the best sites are rainfed maize and soybean areas where production-weighted retention is lowest.

Who bears the risk. A floor defined on the multi-year mean leaves the farmer exposed in the seasons below it: with the density designed for the floor, about 30% of seasons fall short on the main split (70% compliance for SALS), the shortfall is mostly small (2.7% of SALS seasons are below 0.8 of the open-field yield and the 10th percentile of retention is 0.883), and a per-season guarantee through chance-constrained design costs 0.065 LER. Siting by electricity favours maize and soybean areas, where compliance is lowest (Section 3.10), so the risk concentrates where the energy gain is largest. The crop revenue at stake (about 0.3 kUSD ha⁻¹ yr⁻¹) is small against 20–30 kUSD of electricity revenue, so the exposure is mainly one of compliance with the floor and of the contract, and it coincides with the largest uncertainty of the shade response, that of maize and soybean. Who carries it, through contracts, insurance or a premium, is a design question that the model does not address, and food is valued only through crop revenue at market prices.

1164 4.5. Limitations

- 1165 • *Simulation only.* All results come from a simulator. PV output was
1166 benchmarked against PVGIS (29 sites, mean bias -4.9%), but the crop
1167 response to shade was learned from one meta-analysis data set and
1168 compared with four field trials (Sections 3.5 and 3.6). The simulated
1169 response is conservative with respect to the field trials of rice, wheat
1170 and potato; for maize it is conservative with respect to the trials but
1171 milder than the meta-analysis, and for soybean the simulator (0.48 at
1172 40% shading) lies within the interval of the data (0.40, 80% interval
1173 0.34–0.53), while for maize the data give 0.42 against 0.75 in the simu-
1174 lator, so for these crops the direction of the bias depends on the source.
1175 The harvest-index sensitivities h are assumptions, and the sensitivity
1176 analyses use subsamples of crop–site pairs (22 for the rule compari-
1177 son under alternative responses, 21 for the controller comparison, one
1178 training seed).
- 1179 • *Canopy cooling.* The effect of shade on canopy temperature was repre-
1180 sented by one coefficient κ , applied to a heat-stress function that was
1181 calibrated on air temperature (Siebert et al., 2014; Webber et al., 2018).
1182 The ranking of the controllers is robust to κ between 0 and 2°C , but
1183 the absolute electricity changes by about $\pm 8\%$. Nocturnal warming
1184 under modules and rain redistribution are not modelled.
- 1185 • *Crop model.* SIMPLE has a fixed harvest index and linear radiation-
1186 use efficiency, and its CO_2 response is that of controlled experiments
1187 that tend to overstate field responses (Long et al., 2006). More detailed
1188 models with grain-number formation and canopy energy balance may
1189 favour different schedules.
- 1190 • *Layouts and models.* The baseline of static arrays is a backtracking
1191 tracker held at the energy-optimal angle; flat and 30° -rotated (east-
1192 facing, not equator-facing) fixed arrays did no better in a check on
1193 21 pairs, but vertical bifacial layouts (Willockx et al., 2023; Campana
1194 et al., 2021) are not modelled, and the ground coverage ratio is a coarse
1195 proxy of crop productivity (Dupraz, 2023). The PV model is isotropic
1196 in the diffuse component and has no bifacial gain; it was benchmarked
1197 against PVGIS at low density and against pvlib (Holmgren et al., 2018)
1198 at ground coverage ratios up to 0.5, where it underestimates the shading
1199 loss by about 1.5 to 2.5 points; correcting for this changes the LER by

less than 0.007, the break-even prices by less than 1.5 USD MWh⁻¹ and no ranking. The rotation limit was varied between 45 and 80 degrees and changes the size of the benefit but not the ranking; the harvest-index window is assumed to be common to all crops, and shifting it or removing the penalty did not change the ranking of the rules.

- *Climate scenarios.* Warming was applied as uniform temperature shifts with fixed precipitation, radiation and relative humidity, and with unchanged cultivars and sowing dates. It is not a bias-adjusted climate projection.
- *Initialisation and state information.* The network is initialised from the phenology rule, because training from random weights collapsed to full sharing. The gain of SALS is therefore the gain of fine-tuning that rule, and we cannot say whether a better optimiser or a richer policy class would find larger gains. The controller uses the simulated state of the crop and soil (thermal time, canopy interception, soil water) without measurement error, and the forecast errors we tested apply to weather only.
- *Economics and deployment.* The economic analysis (Section 3.8) uses assumed prices and costs, no grid, curtailment, storage or subsidy model, and does not value food security; its break-even prices move with the capex and the discount rate (Fig. 7). The value of midday electricity, curtailment, storage, structural cost and farmer acceptance (Trommsdorff et al., 2021) will determine the realisable share. The average row pitch at the chosen densities (about 13 m for 2.2-m modules) leaves room for machinery, but the densest designs (GCR up to 0.5) have a pitch of only 4.4 m.
- *Floor and field effects.* The floor of 0.9 is a modelling convention and not a regulatory threshold (DIN SPEC 91434 uses 66%, Supplementary Material). No field validation under trackers exists for the controllers, and the simulator has no emergence, early canopy closure, weed or machinery effects; the PV model has not been compared with measured array output at ground coverage above 0.05, only with pvlib.

5. Conclusions

Light sharing, the daily choice between maximum electricity and letting light through to the crop, is the main lever on the electricity of tracker-based agrivoltaics at a given yield-retention floor. In a differentiable simulator, evaluated on held-out crop–site pairs (92 on the main split, 76 distinct pairs pooled over three splits for the controller and design comparisons) with the density of each site chosen from its own history, light-sharing rules allowed 2.1 to 2.9 times the module density of static arrays at a mean yield retention of 90% and raised the land-equivalent ratio from 1.07 to 1.18–1.24 under the proportional-growth shade response. A rule that follows crop phenology and needs no weather forecast was the best written rule. The lever pays only at high electricity prices: with illustrative costs it breaks even with the open field, at floor-designed densities, at 70 to 92 USD MWh^{−1} (73 for the learned controller, 59 for static arrays), static arrays earn more than the light-sharing methods below about 75 to 85 USD MWh^{−1} (the first value for profit-optimal densities, the second for floor-designed ones), and a conventional plant replacing the crop is more profitable above its break-even of 48 USD MWh^{−1}. In this model agrivoltaics therefore pays only with a premium, land-use rules or a value for food production that the model does not include.

The crop’s response to shade, the least certain input, was learned from 428 field observations. For the staple crop types a hierarchical model predicted a held-out study no better than proportional loss (error 0.22), which measures the between-study spread, and gradient descent through the simulator turned the learned curves and four field trials into ensembles of plausible responses. The simulator is conservative against the trials for rice, wheat and potato; for maize and soybean, which supply 53% of the upscaled electricity, the meta-analysis indicates larger losses than simulated for maize, whereas for soybean the simulated response lies within the interval of the data, and one maize trial shows smaller losses; the gains for these crops are therefore uncertain. A learned controller exceeded the written phenology rule by 0.044 LER but a CMA-ES-optimised window by only 0.007 (95% interval 0.002–0.014, mean of three training seeds per split; at matched retention 0.003, interval −0.0004 to 0.007). The source of this margin is not identified: it is not explained by the tuning of a window or a ramp, and masking soil water, aridity and forecasts leaves the controller unchanged. Training on ensembles of shade responses, or giving the controller a farm-measurable biomass signal, did not help within our budget (one seed, 21 pairs).

1269 Compliance depends more on the response assumed in the design than
 1270 on the controller. A density chosen under the mean response left about 70%
 1271 of crop–site pairs at the floor when another plausible response held, against
 1272 85–87% for an oracle design; this gap of 15 points (a model-misspecification
 1273 effect) is larger than the 4 points between light-sharing controllers at a fixed
 1274 design rule under a known response, but the two are not like-for-like. Design-
 1275 ing for the floor in 75% of the plausible responses recovered about half of the
 1276 gap at a cost of 0.03 LER, and a floor raised by 0.02 under the mean response
 1277 gave no detectable difference at equal compliance (the intervals include zero
 1278 and are too wide to establish equivalence); no rule was safe outside the en-
 1279 semble, whose members mostly differ less from one another than a new study
 1280 differs from the learned curve (not for soybean under the widest ensemble).
 1281 The ensemble is therefore useful to size a safety margin, not to replace it.
 1282 The yield floor is met on average but not in every season or region, and the
 1283 electricity potential of 5% of staple cropland (about 14 PWh yr⁻¹, a scaling
 1284 convention without an interval) requires about 10 TWp of modules. Future
 1285 work should validate the crop response to shade under trackers with more
 1286 field data, notably for maize and legumes where the meta-analysis and the
 1287 trials disagree, extend the crop model to grain-number formation and canopy
 1288 energy balance, add costs and grid constraints, and pursue controllers that
 1289 guarantee the yield-retention floor per site and per season.

1290 **CRedit authorship contribution statement**

1291 **Quang-Vinh Dang:** Conceptualization, Methodology, Software, Formal
 1292 analysis, Investigation, Data curation, Visualization, Writing – original draft,
 1293 Writing – review & editing.

1294 **Declaration of competing interest**

1295 The author has no known competing financial interests or personal re-
 1296 lationships that could have appeared to influence the work reported in this
 1297 paper.

1298 **Data and code availability**

1299 All data used are public: SPAM 2010 v2r0 harvested areas (Yu et al.,
 1300 2020), the GGCM Phase 3 crop calendar (Jägermeyr et al., 2021), NASA

1301 POWER daily weather (Sparks, 2018) the SIMPLE crop parameters (Zhao
 1302 et al., 2019) and the shade-response meta-analysis data of Laub et al. (2022)
 1303 (Zenodo 10.5281/zenodo.5716091, CC BY 4.0, copied into the repository).
 1304 The code of the simulator, of SALS and of all experiments, together with the
 1305 site list, is available from the author’s public repository ([https://github.](https://github.com/vinhqddang/Artificial-Intelligence-in-Agriculture)
 1306 [com/vinhqddang/Artificial-Intelligence-in-Agriculture](https://github.com/vinhqddang/Artificial-Intelligence-in-Agriculture)); a versioned
 1307 archive with a DOI will be deposited at acceptance.

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Table 2: Held-out performance (92 test sites, seasons 2015–2019, canopy cooling $\kappa = 1^\circ\text{C}$). The density (GCR) of each site is the largest value on the grid 0.05, 0.075, ..., 0.5 at which the method’s mean retention in the site’s 2001–2014 seasons is ≥ 0.9 . E : electricity ($\text{MWh ha}^{-1} \text{ yr}^{-1}$); e : electricity relative to a conventional PV plant; r : mean yield retention; r_{prod} : production-weighted retention; C_s and C_{site} : share of seasons with $r \geq 0.9$ and share of sites whose test-season mean $r \geq 0.9$ (%); $\text{LER} = r + e$.

Method	GCR	E	e	r	r_{prod}	C_s	C_{site}	LER
<i>Baseline</i>								
AV static	0.06	217	0.16	0.917	0.914	93	96	1.07
Seasonal sharing	0.17	397	0.28	0.904	0.902	69	72	1.19
Phenology rule	0.17	465	0.33	0.904	0.902	69	75	1.24
Stress rule	0.13	387	0.28	0.907	0.907	76	82	1.18
SALS	0.17	510	0.36	0.905	0.902	70	74	1.27
<i>+1.5 °C</i>								
AV static	0.06	221	0.16	0.916	0.914	93	92	1.08
Seasonal sharing	0.16	393	0.28	0.905	0.903	72	74	1.19
Phenology rule	0.17	478	0.34	0.905	0.902	68	75	1.25
Stress rule	0.12	377	0.27	0.908	0.908	73	77	1.18
SALS	0.17	518	0.37	0.906	0.903	68	75	1.28
<i>+2 °C</i>								
AV static	0.06	223	0.16	0.916	0.914	92	93	1.08
Seasonal sharing	0.17	394	0.28	0.905	0.904	72	76	1.19
Phenology rule	0.17	483	0.35	0.905	0.903	69	74	1.25
Stress rule	0.12	370	0.26	0.909	0.909	75	78	1.17
SALS	0.17	521	0.38	0.907	0.904	69	75	1.28
<i>+3 °C</i>								
AV static	0.06	231	0.17	0.915	0.914	88	89	1.08
Seasonal sharing	0.17	397	0.29	0.906	0.904	69	75	1.19
Phenology rule	0.18	499	0.36	0.905	0.902	67	72	1.26
Stress rule	0.11	366	0.26	0.908	0.908	74	77	1.17
SALS	0.18	540	0.39	0.907	0.904	68	75	1.30

Table 3: Rules and the learned controller (SALS: mean of three training seeds per split) pooled over the three spatial splits (76 distinct crop–site pairs, every third held-out pair; baseline climate; main shade response; history-based design on a GCR grid with step 0.05, coarser than the 0.025 grid of the main analysis). The grid-tuned window shares light while thermal time lies between 0.15 and 1.0 of the season requirement (best of a 4×3 grid on the training-block seasons of each split, the same on all three; the written rule uses 0.05 to 1.0). The CMA-ES schedule has five parameters (start, end, two ramp widths, level) was tuned on each split’s training block and stopped, at the generation cap, at an almost binary window from 0.20 to 0.93 of the requirement at level 1.00 (see Methods).

Method	GCR	E/E_{PV}	Retention	Seasons ≥ 0.9 (%)	Pairs ≥ 0.9 (%)	LER
AV static	0.059	0.153	0.922	95	96	1.076
Seasonal sharing	0.151	0.249	0.915	84	91	1.164
Stress rule	0.118	0.255	0.918	83	88	1.173
Phenology rule (written)	0.154	0.292	0.916	85	91	1.207
Grid-tuned window	0.161	0.317	0.916	84	89	1.233
CMA-ES optimised window	0.158	0.327	0.916	83	87	1.244
SALS	0.161	0.335	0.916	83	87	1.251

Table 4: Simulated relative yield at the shading of published field trials (static arrays, training-block sites). Observed values: rice, single trial; wheat and potato, the worse of the two seasons (the better season gave +2.7% and +11%); maize, mean of the three seasons of the dynamic system (DAV) and single values for the other two systems. The same trials enter the calibration of Section 2.6, so the comparison is not independent.

Trial	Shading	Observed	Conservative	Calibrated
Rice, Gonocruz et al. (2021)	0.25	0.84	0.77	0.79
Wheat, Weselek et al. (2021)	0.30	0.81	0.66	0.74
Potato, Weselek et al. (2021)	0.30	0.82	0.72	0.72
Maize, Ramos-Fuentes et al. (2023) (DAV)	0.35	0.83	0.70	0.70
Maize, Ramos-Fuentes et al. (2023) (half)	0.30	0.91	0.75	0.75
Maize, Ramos-Fuentes et al. (2023) (full)	0.50	0.70	0.54	0.54

Table 5: Relative yield at 20% and 40% shading: data-driven model (median and 80% bootstrap interval over studies; crop-type curve, covariates at their reference values: mean irradiation, temperate zone, no solar-panel shade, no intercropping) and simulator under the two calibrated ensembles (mean of members; E1 fitted to hand-read curves, E2 fitted to the data-driven curves). The data-driven curve of rice is that of C3 cereals (no rice in the data set). Observations and studies: C3 cereals 41/10, maize 51/10, grain legumes 76/14, tubers 12/2.

Crop	Shading	Data-driven	Simulator, E1	Simulator, E2	Proportional
Wheat	20%	0.93 [0.88, 0.96]	0.88	0.92	0.80
Wheat	40%	0.72 [0.59, 0.78]	0.70	0.77	0.60
Rice	20%	0.93 [0.88, 0.96]	0.85	0.90	0.80
Rice	40%	0.72 [0.59, 0.78]	0.66	0.73	0.60
Maize	20%	0.73 [0.58, 0.86]	0.86	0.90	0.80
Maize	40%	0.42 [0.27, 0.59]	0.69	0.75	0.60
Soybean	20%	0.71 [0.65, 0.79]	0.77	0.74	0.80
Soybean	40%	0.40 [0.34, 0.53]	0.53	0.48	0.60
Potato	20%	0.93 [0.92, 0.97]	0.88	0.92	0.80
Potato	40%	0.71 [0.67, 0.83]	0.73	0.79	0.60

Table 6: Land-equivalent ratio of rules and learned controllers under four true shade responses, with the array density designed under the true response (split 0, baseline climate, floor $\rho = 0.9$, every sixth held-out pair, 21 pairs, coarse GCR grid, one training seed; season means, so differences differ slightly from the pair-level intervals in the text). SALS is trained on the proportional response; robust controllers are trained on random members of ensemble E1, E2 or E3 (adaptive: with the measured biomass ratio). The proportional and harsh responses lie outside the ensembles.

Method	Proportional	Calibrated	Field-fitted	Harsh
AV static	1.052	1.059	1.254	1.037
Seasonal sharing	1.133	1.156	1.385	1.097
Stress rule	1.129	1.160	1.406	1.086
Feedback rule	1.139	1.164	1.392	1.089
Phenology rule (written)	1.164	1.190	1.456	1.122
Grid-tuned window	1.170	1.199	1.472	1.129
CMA-ES optimised window	1.173	1.198	1.479	1.125
SALS	1.178	1.209	1.489	1.132
SALS-robust, E1	1.179	1.204	1.491	1.127
SALS-adaptive, E1	1.176	1.202	1.495	1.124
SALS-robust, E2	1.165	1.192	1.483	1.126
SALS-robust, E3	1.163	1.191	1.481	1.121
SALS-adaptive, E3	1.164	1.191	1.475	1.118

Table 7: Array design under shade-response uncertainty, phenology rule, three spatial splits pooled (76 distinct crop-site pairs, every third held-out pair). Truth: each of five members of the ensemble in turn; the mean response and the member-based rules use the other four members (leave-one-out, so the truth never enters the design). E2: ensemble calibrated to the learned curves and field trials; E3: same procedure with target errors of 0.2 on the learned curves. Pairs: share with mean retention ≥ 0.9 ; shortfall: mean of $\max(0, 0.9 - \text{retention})$ per pair.

Design	E2		LER	E3		
	Pairs ≥ 0.9 (%)	Shortfall		Pairs ≥ 0.9 (%)	Shortfall	
oracle (true response)	87	0.0025	1.486	85	0.0029	1
mean response	70	0.0062	1.493	71	0.0093	1
floor met in 50% of other members	65	0.0084	1.510	67	0.0125	1
floor met in 75% of other members	78	0.0045	1.462	79	0.0083	1
floor met in all other members	89	0.0019	1.412	88	0.0032	1
floor 0.92 under mean response	88	0.0022	1.426	85	0.0051	1
floor 0.94 under mean response	97	0.0007	1.345	92	0.0027	1
floor 0.96 under mean response	99	0.0003	1.251	96	0.0015	1

Table 8: Design rules outside the ensemble: share of crop–site pairs (%) with mean retention ≥ 0.9 under four fixed responses (P proportional, C calibrated to the meta-analysis, F field-fitted, H harsh), phenology rule, three splits pooled (76 distinct pairs). The member-based rules use the five members of E2 or E3, the mean-response design and the margins use the mean of all twelve members, and the oracle uses the true response. LER in the last two columns is the mean over the four responses; the oracle rows are identical for E2 and E3 because the oracle uses the true response.

Design	E2				E3				LER	
	P	C	F	H	P	C	F	H	E2	E3
oracle (true response)	91	89	87	84	91	89	87	84	1.283	1.283
mean response	13	13	68	12	13	17	79	7	1.435	1.448
floor met in 75% of members	16	16	80	12	22	25	93	13	1.384	1.397
floor met in all members	17	17	80	12	29	32	93	16	1.348	1.336
floor 0.92 under mean response	17	20	86	12	29	33	91	20	1.376	1.399
floor 0.94 under mean response	45	47	97	34	45	45	97	29	1.303	1.345
floor 0.96 under mean response	63	78	99	54	58	63	99	51	1.220	1.264

Table 9: Assumed economic parameters (illustrative, not data). Ranges used in the sensitivity analysis are in parentheses.

Parameter	Value
Electricity price	60 USD MWh ⁻¹ (30–120)
Capex, AV single-axis tracker plant	1.0 USD Wp ⁻¹ (0.7–1.5)
Capex, conventional tracker plant	0.75 USD Wp ⁻¹
Discount rate, lifetime, O&M	6% (4–8%), 25 years, 1.5% of capex per year
Installed capacity	2.1 MWp per hectare of module area share g (0.21 kWp m ⁻²)
Crop price per tonne of dry matter	wheat 290, rice 400, maize 250, soybean 520, potato 1000 USD (0.

Table 10: Illustrative annual economics per hectare (76 crop–site pairs, three splits, main shade response, baseline climate; parameters in Table 9). Net value is relative to the open field (crop costs unchanged) in kUSD ha⁻¹ yr⁻¹ at three electricity prices; break-even is the price at which the net value is zero. The conventional plant replaces the crop.

Method	E (MWh)	Capex annuity + O&M	Crop loss	Net at 60	Net at 90
AV static	200	11.5	0.27	0.3	6.3
Seasonal sharing	326	29.6	0.31	-10.3	-0.5
Stress rule	336	23.2	0.30	-3.3	6.8
Phenology rule (written)	382	30.1	0.30	-7.5	4.0
Grid-tuned window	415	31.6	0.31	-7.0	5.5
CMA-ES optimised window	428	31.0	0.30	-5.6	7.3
SALS	438	31.6	0.31	-5.6	7.6
Conventional PV plant	1306	58.7	3.60	16.0	55.2